MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INTERFACE, MICROPROCESSOR, KEY CODE NGK

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic microprocessor interface module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.1024 failures per million operating hours for class I or 0.2182 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be NGK.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/328-1 for class I and M28787/328-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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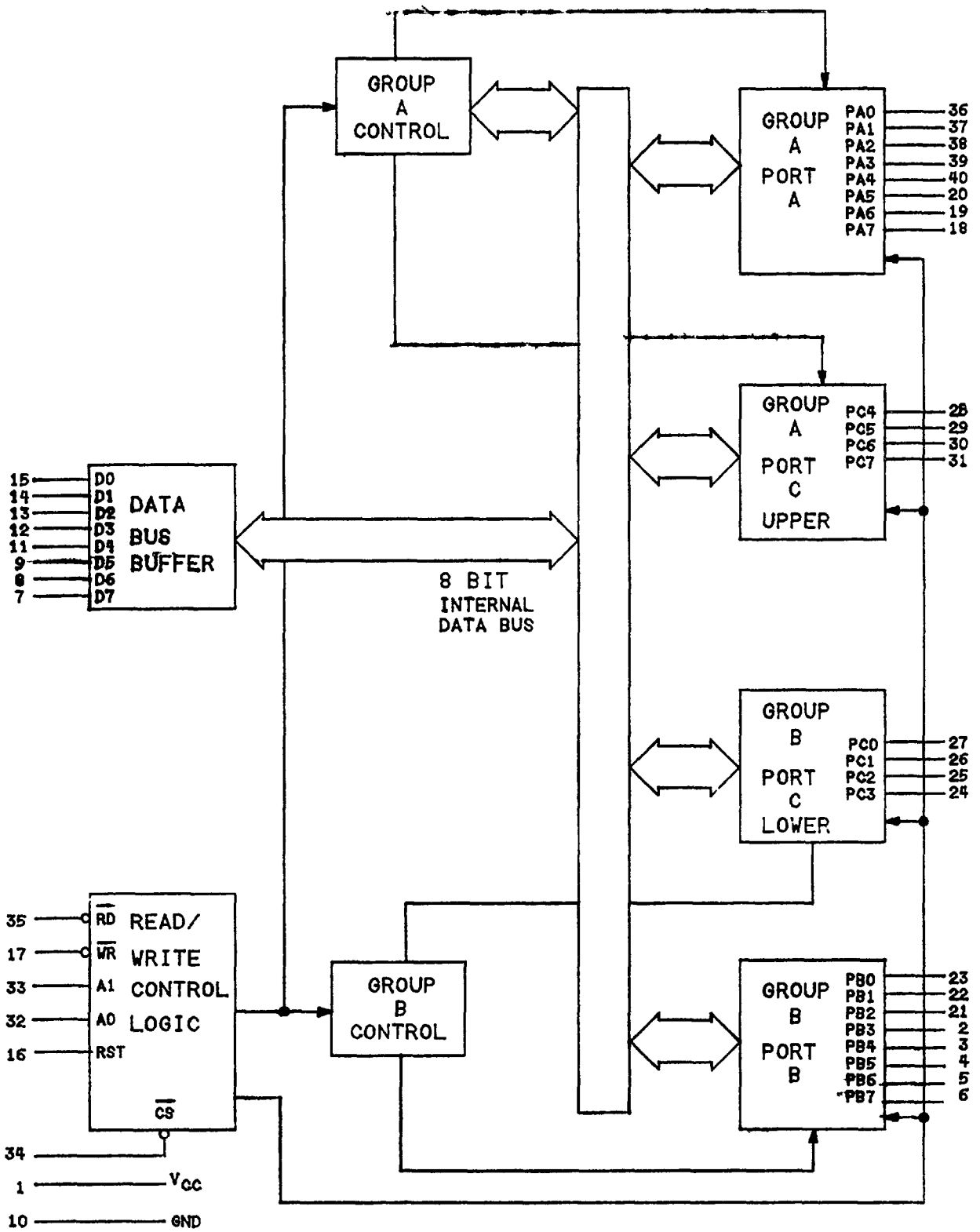


FIGURE 1. NGK module function diagram, microprocessor interface.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} stress	III	S	← Must pass remaining tests →										
C1-C32	Output short circuit check	IV	S	← Must pass remaining tests →										
D1-D38	Input current, low level	V	X	0	-10	0	-10	0	-10	0	-10	0	-10	µA
E1-E38	Input current, high level	VI	X	0	10	0	10	0	10	0	10	0	10	µA
F1-F32	Output voltage, low level	VII	X	0	450	0	450	0	450	0	450	0	450	mV
G1-G32	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H16	Output current, high level	IX	X	1	4	1	4	1	4	1	4	1	4	mA
J1,J2	Dynamic functional	XII	X	← See table XII →										
K1-K8	Disable delay time (t_{PHZ})	XIV	S	21	210	21	210	21	210	21	210	21	210	ns
K9-K16	Disable delay time (t_{PHZ})	XIV	S	40	400	40	400	40	400	40	400	40	400	ns
L1-L8	Disable delay time (t_{PLZ})	XV	S	10	150	10	150	10	150	10	150	10	150	ns
L9-L16	Disable delay time (t_{PLZ})	XV	S	20	300	20	300	20	300	20	300	20	300	ns
M	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
N1-N32	Output crosstalk, low level	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
P1-P32	Output crosstalk, high level	XXI	Q	0	400	0	400	0	400	0	400	0	400	mV
Q	Power supply filter capacitance	XXIII	S	1.5	2.9	1.5	2.9	1.5	2.9	1.5	2.9	1.5	2.9	µF

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R32	Input capacitance	XXIV	Q	2	25	2	25	2	25	2	25	2	25	pF
R33-R38	Input capacitance	XXIV	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through Q shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Q, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through R shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	660	660	mW
Input voltage			
Low level	800 max	800 max	mV
High level	2 min	2 min	V

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	Class I	Class II	Unit
Input current			
Low level (V_{IL} = 0 V)	-10 max	-10 max	µA
High level (V_{IH} = V_{CC})	10 max	10 max	µA
Output drive current			
Low level (V_{OL} = 450 mV)	1.7 max	1.7 max	mA
High level			
V_{OH} = 2.4 V (D0-D7)	-400 max	-400 max	µA
(PA,PB,PC)	-200 max	-200 max	µA
V_{OH} = 1.5 V (PB,PC)	-4 max	-4 max	mA
Setup and hold time	See table XII		

6.5 Module function. This module contains an 8-bit programmable interface for a microprocessor.

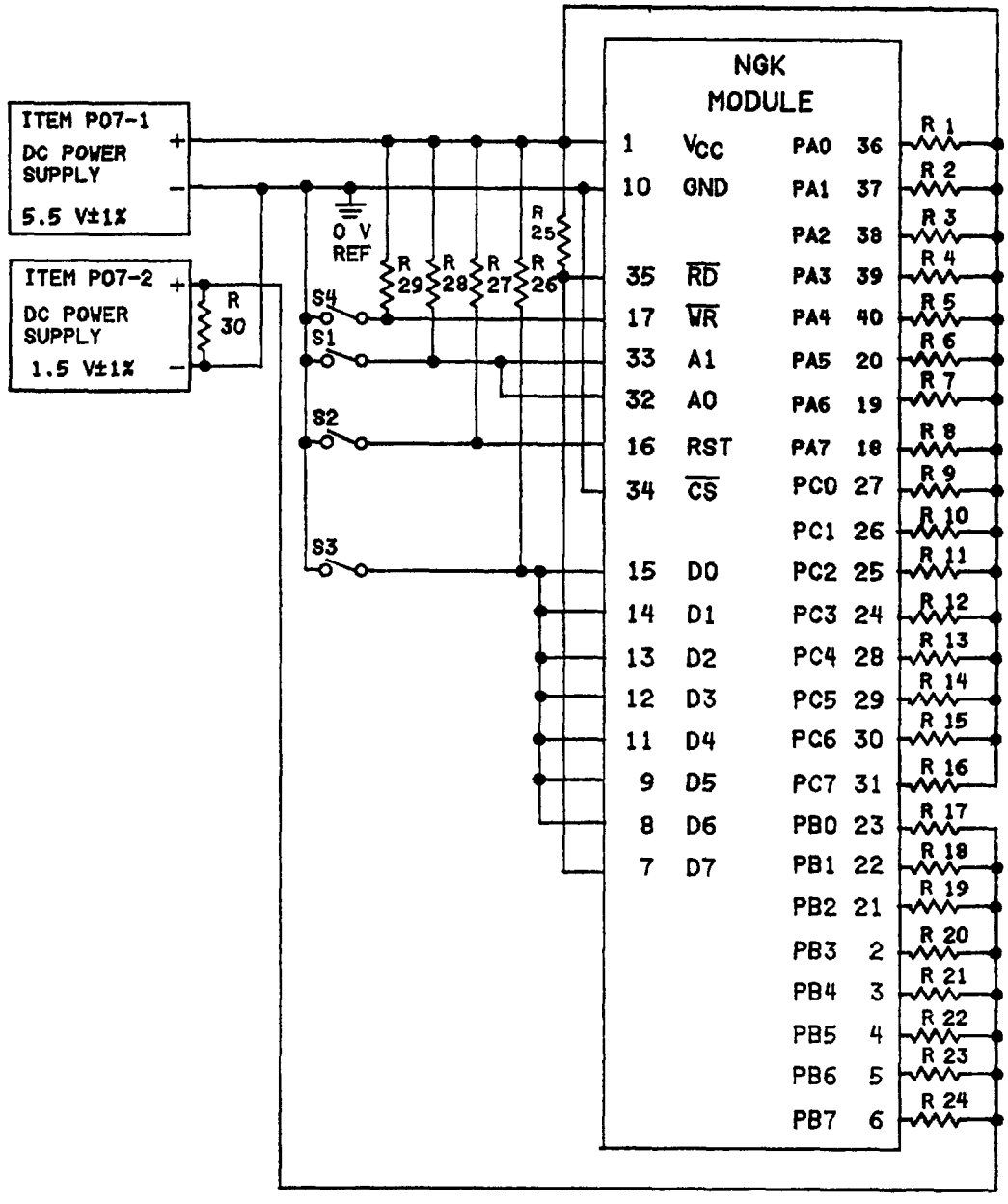
6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and may be obtained upon request.

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NOTES:
1. R1-R16 = 25.3 kΩ ±1%, 1/8 W.
2. R17-R24 = 750Ω ±1%, 1/8 W.
3. R25-R29 = 1 kΩ ±10%, 1/8 W.
4. R30 = 51.1Ω ±1%, 1/2 W.
5. S1-S4 = SPST switch.
6. Initialization sequence:

	S1	S2	S3	S4
1	open	open	open	closed
2	open	closed	closed	open
3	closed	closed	open	closed
4	closed	closed	open	open

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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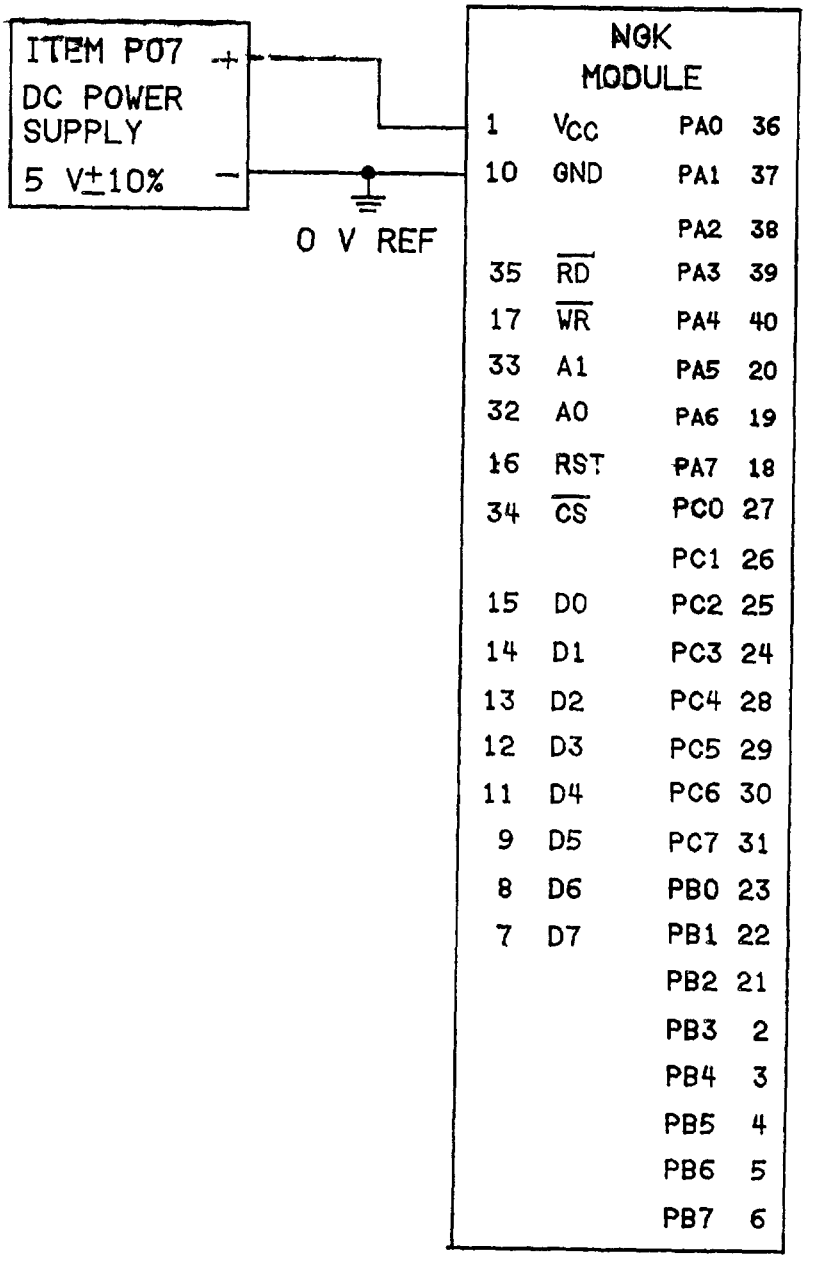


FIGURE 3. Vibration test circuit.

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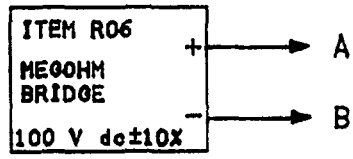
TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	4

TABLE III. V_{CC} stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
B	V_{CC} stress	Module must pass remaining tests			5

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NGK MODULE			
1	V_{CC}	PA0	36
10	GND	PA1	37
		PA2	38
35	$\overline{RD}$	PA3	39
17	$\overline{WR}$	PA4	40
33	A1	PA5	20
32	A0	PA6	19
16	RST	PA7	18
34	$\overline{CS}$	PC0	27
		PC1	26
15	D0	PC2	25
14	D1	PC3	24
13	D2	PC4	28
12	D3	PC5	29
11	D4	PC6	30
9	D5	PC7	31
8	D6	PB0	23
7	D7	PB1	22
		PB2	21
		PB3	2
		PB4	3
		PB5	4
		PB6	5
		PB7	6

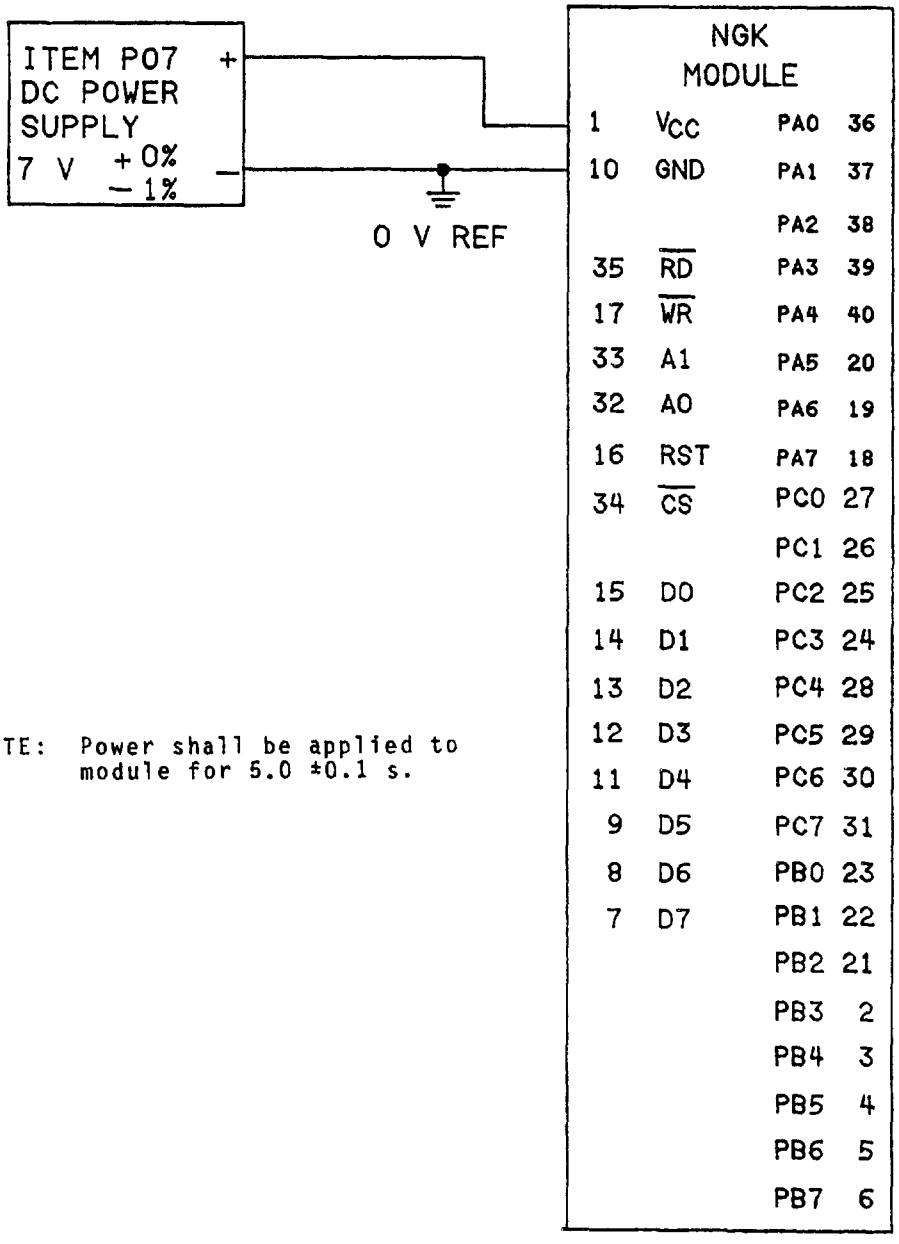
WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative lead.

FIGURE 4. Isolation test circuit for table II.

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NOTE: Power shall be applied to module for 5.0 ±0.1 s.

FIGURE 5. V_{CC} stress test circuit for table III.

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TABLE IV. Output short circuit check test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Test figure
			A to pin 1/	Min	Max	
C1	PA0	Output short circuit check	36	2/	2/	6
C2	PA1	"	37	"	"	"
C3	PA2	"	38	"	"	"
C4	PA3	"	39	"	"	"
C5	PA4	"	40	"	"	"
C6	PA5	"	20	"	"	"
C7	PA6	"	19	"	"	"
C8	PA7	"	18	"	"	"
C9	PB0	"	23	"	"	"
C10	PB1	"	22	"	"	"
C11	PB2	"	21	"	"	"
C12	PB3	"	2	"	"	"
C13	PB4	"	3	"	"	"
C14	PB5	"	4	"	"	"
C15	PB6	"	5	"	"	"
C16	PB7	"	6	"	"	"
C17	PC0	"	27	"	"	"
C18	PC1	"	26	"	"	"
C19	PC2	"	25	"	"	"
C20	PC3	"	24	"	"	"
C21	PC4	"	28	"	"	"

See footnotes at end of table.

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TABLE IV. Output short circuit check test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Test figure
		A to pin 1/	Min	Max	
C22	PC5 Output short circuit check	29	2/	2/	6
C23	PC6 "	30	"	"	"
C24	PC7 "	31	"	"	"
C25	D0 "	15	"	"	"
C26	D1 "	14	"	"	"
C27	D2 "	13	"	"	"
C28	D3 "	12	"	"	"
C29	D4 "	11	"	"	"
C30	D5 "	9	"	"	"
C31	D6 "	8	"	"	"
C32	D7 "	7	"	"	"

1/ Connect one output at a time after reaching the test point from table XI. Do not exceed one second each.
2/ Module must pass all remaining tests.

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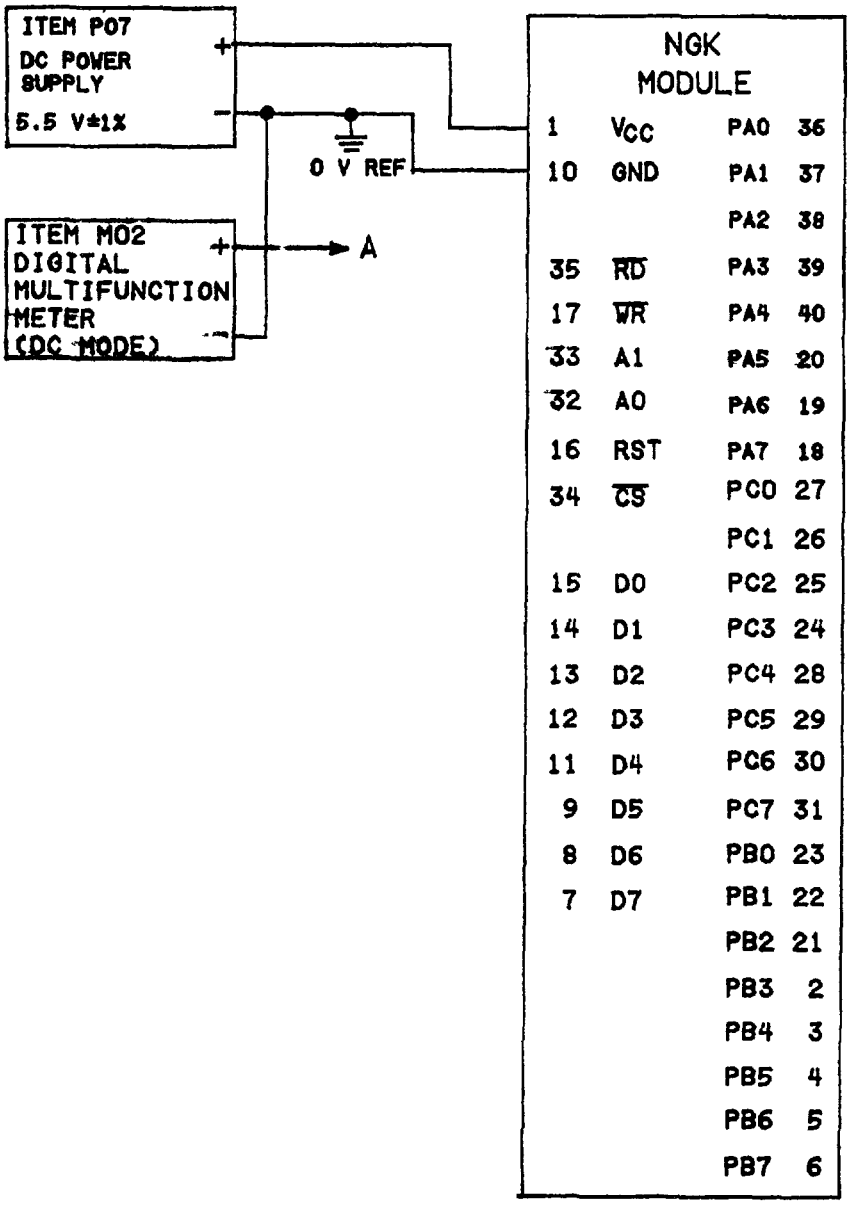


FIGURE 6. Output short circuit current check test circuit for table IV.

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TABLE V. Input current (low-level) test.

Test condition	Parameter		Connect 1/				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	D0	Input current, low level	35,14,13,12, 11,9,8,7	15	16,17,32,33, 34	Open	0	-10	µA	7
D2	D1	"	35,15,13,12, 11,9,8,7	14	"	"	"	"	"	"
D3	D2	"	35,15,14,12, 11,9,8,7	13	"	"	"	"	"	"
D4	D3	"	35,15,14,13, 11,9,8,7	12	"	"	"	"	"	"
D5	D4	"	35,15,14,13, 12,9,8,7	11	"	"	"	"	"	"
D6	D5	"	35,15,14,13, 12,11,8,7	9	"	"	"	"	"	"
D7	D6	"	35,15,14,13, 12,11,9,7	8	"	"	"	"	"	"
D8	D7	"	35,15,14,13, 12,11,9,8	7	"	"	"	"	"	"
D9	PA0	"	16,37,38,39, 40,20,19,18	36	34	"	"	"	"	"
D10	PA1	"	16,36,38,39, 40,20,19,18	37	"	"	"	"	"	"
D11	PA2	"	16,36,37,39, 40,20,19,18	38	"	"	"	"	"	"
D12	PA3	"	16,36,37,38, 40,20,19,18	39	"	"	"	"	"	"
D13	PA4	"	16,36,37,38, 39,20,19,18	40	"	"	"	"	"	"
D14	PA5	"	16,36,37,38, 39,40,19,18	20	"	"	"	"	"	"

See footnote at end of table.

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TABLE V. Input current (low-level) test - Continued.

Test condition	Parameter	Connect 1/				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D15	PA6 Input current, low level	16,36,37,38, 39,40,20,18	19	34	Open	0	-10	µA	7
D16	PA7 "	16,36,37,38, 39,40,20,19	18	"	"	"	"	"	"
D17	PB0 "	16,22,21,2,3, 4,5,6	23	"	"	"	"	"	"
D18	PB1 "	16,23,21,2,3, 4,5,6	22	"	"	"	"	"	"
D19	PB2 "	16,23,22,2,3, 4,5,6	21	"	"	"	"	"	"
D20	PB3 "	16,23,22,21,3, 4,5,6	2	"	"	"	"	"	"
D21	PB4 "	16,23,22,21,2, 4,5,6	3	"	"	"	"	"	"
D22	PB5 "	16,23,22,21,2, 3,5,6	4	"	"	"	"	"	"
D23	PB6 "	16,23,22,21,2, 3,4,6	5	"	"	"	"	"	"
D24	PB7 "	16,23,22,21,2, 3,4,5	6	"	"	"	"	"	"
D25	PC0 "	16,26,25,24, 28,29,30,31	27	"	"	"	"	"	"
D26	PC1 "	16,27,25,24, 28,29,30,31	26	"	"	"	"	"	"
D27	PC2 "	16,27,26,24, 28,29,30,31	25	"	"	"	"	"	"
D28	PC3 "	16,27,26,25, 28,29,30,31	24	"	"	"	"	"	"

See footnote at end of table.

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TABLE V. Input current (low-level) test - Continued.

Test condition	Parameter	Connect 1/				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D29	PC4 Input current, low level	16,27,26,25, 24,29,30,31	28	34	Open	0	-10	µA	7
D30	PC5 "	16,27,26,25, 24,28,30,31	29	"	"	"	"	"	"
D31	PC6 "	16,27,26,25, 24,28,29,31	30	"	"	"	"	"	"
D32	PC7 "	16,27,26,25, 24,28,29,30	31	"	"	"	"	"	"
D33	$\overline{RD}$ "	17,33,32	35	"	"	"	"	"	"
D34	$\overline{WR}$ "	35,33,32	17	"	"	"	"	"	"
D35	A1 "	35,17,32	33	"	"	"	"	"	"
D36	A0 "	35,17,33	32	"	"	"	"	"	"
D37	RST "	35,17,33,32	16	"	"	"	"	"	"
D38	$\overline{CS}$ "	35,17,33,32,16	34	Open	"	"	"	"	"

1/ Before making measurements, toggle pin 16 high then low.

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TABLE VI. Input current (high-level) test.

Test condition	Parameter	Connect 1/				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	DO Input current, high level	35	Open	17,32,33,34,14,13, 12,11,9,8,7	15	0	10	µA	7
E2	D1 "	"	"	17,32,33,34,15,13, 12,11,9,8,7	14	"	"	"	"
E3	D2 "	"	"	17,32,33,34,15,12, 11,9,8,7	13	"	"	"	"
E4	D3 "	"	"	17,32,33,34,15,14, 11,9,8,7	12	"	"	"	"
E5	D4 "	"	"	17,32,33,34,15,14, 13,12,9,8,7	11	"	"	"	"
E6	D5 "	"	"	17,32,33,34,15,14, 13,12,11,8,7	9	"	"	"	"
E7	D6 "	"	"	17,32,33,34,15,14, 13,12,11,9,7	8	"	"	"	"
E8	D7 "	"	"	17,32,33,34,15,14, 13,12,11,9,8	7	"	"	"	"
E9	PA0 "	16	"	34,37,38,39,40,20, 19,18	36	"	"	"	"
E10	PA1 "	"	"	34,36,38,39,40,20, 19,18	37	"	"	"	"
E11	PA2 "	"	"	34,36,37,39,40,20, 19,18	38	"	"	"	"
E12	PA3 "	"	"	34,36,37,38,40,20, 19,18	39	"	"	"	"
E13	PA4 "	"	"	34,36,37,38,39,20, 19,18	40	"	"	"	"
E14	PA5 "	"	"	34,36,37,38,39,40, 19,18	20	"	"	"	"

See footnote at end of table.

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TABLE VI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect 1/				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E15	PA6 Input current, high level	16	Open	34,36,37,38,39,40, 20,18	19	0	10	µA	7
E16	PA7 "	"	"	34,36,37,38,39,40, 20,19	18	"	"	"	"
E17	PB0 "	"	"	34,22,21,2,3,4, 5,6	23	"	"	"	"
E18	PB1 "	"	"	34,23,21,2,3,4, 5,6	22	"	"	"	"
E19	PB2 "	"	"	34,23,22,2,3,4, 5,6	21	"	"	"	"
E20	PB3 "	"	"	34,23,22,21,3,4, 5,6	2	"	"	"	"
E21	PB4 "	"	"	34,23,22,21,2,4, 5,6	3	"	"	"	"
E22	PB5 "	"	"	34,23,22,21,2,3, 5,6	4	"	"	"	"
E23	PB6 "	"	"	34,23,22,21,2,3, 4,6	5	"	"	"	"
E24	PB7 "	"	"	34,23,22,21,2,3, 4,5	6	"	"	"	"
E25	PC0 "	"	"	34,26,25,24,28,29, 30,31	27	"	"	"	"
E26	PC1 "	"	"	34,27,25,24,28,29, 30,31	26	"	"	"	"
E27	PC2 "	"	"	34,27,26,24,28,29, 30,31	25	"	"	"	"
E28	PC3 "	"	"	34,27,26,25,28,29, 30,31	24	"	"	"	"

See footnote at end of table.

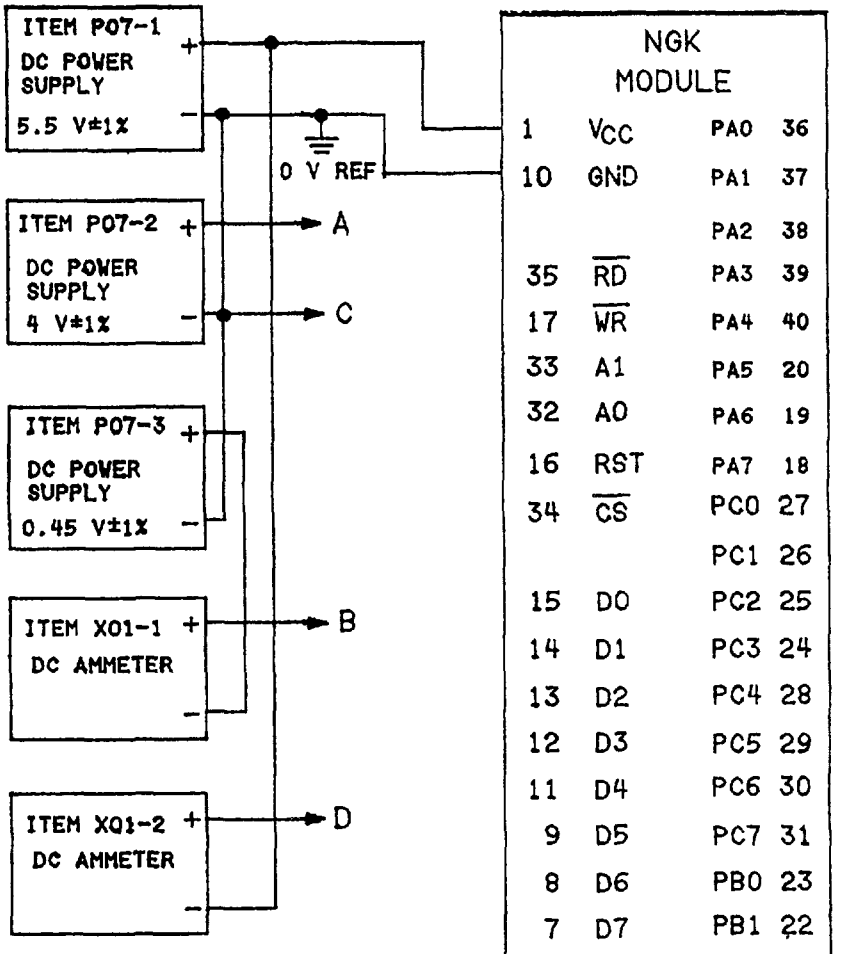
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TABLE VI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect 1/				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E29	PC4 Input current, high level	16	Open	34,27,26,25,24,29, 30,31	28	0	10	µA	7
E30	PC5 "	"	"	34,27,26,25,24,28, 30,31	29	"	"	"	"
E31	PC6 "	"	"	34,27,26,25,24,28, 29,31	30	"	"	"	"
E32	PC7 "	"	"	34,27,26,25,24,28, 29,30	31	"	"	"	"
E33	$\overline{RD}$ "	Open	"	17,33,32,16,34	35	"	"	"	"
E34	$\overline{WR}$ "	"	"	35,33,32,16,34	17	"	"	"	"
E35	A1 "	"	"	35,17,32,16,34	33	"	"	"	"
E36	A0 "	"	"	35,17,33,16,34	32	"	"	"	"
E37	RST "	"	"	35,17,33,32,34	16	"	"	"	"
E38	$\overline{CS}$ "	"	"	35,17,33,32,16	34	"	"	"	"

1/ Before making measurements, toggle pin 16 high then low.

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CAUTION: Do not isolate the chassis of item X01 from earth ground.

NOTE: Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

FIGURE 7. Input current (low-level and high-level) test circuit for tables V and VI.

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TABLE VII. Output voltage (low-level) test.

Test condition	Parameter	Connect 1/	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F1	BA0 Output voltage, low level	36	0	450	mV	8
F2	BA1 "	37	"	"	"	"
F3	BA2 "	38	"	"	"	"
F4	BA3 "	39	"	"	"	"
F5	BA4 "	40	"	"	"	"
F6	BA5 "	20	"	"	"	"
F7	BA6 "	19	"	"	"	"
F8	BA7 "	18	"	"	"	"
F9	PB0 "	23	"	"	"	"
F10	PB1 "	22	"	"	"	"
F11	PB2 "	21	"	"	"	"
F12	PB3 "	2	"	"	"	"
F13	PB4 "	3	"	"	"	"
F14	PB5 "	4	"	"	"	"
F15	PB6 "	5	"	"	"	"
F16	PB7 "	6	"	"	"	"
F17	PC0 "	27	"	"	"	"
F18	PC1 "	26	"	"	"	"
F19	PC2 "	25	"	"	"	"
F20	PC3 "	24	"	"	"	"
F21	PC4 "	28	"	"	"	"

See footnote at end of table.

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TABLE VII. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect 1/	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F22	PC5 Output voltage, low level	29	0	450	mV	8
F23	PC6 "	30	"	"	"	"
F24	PC7 "	31	"	"	"	"
F25	D0 "	15	"	"	"	"
F26	D1 "	14	"	"	"	"
F27	D2 "	13	"	"	"	"
F28	D3 "	12	"	"	"	"
F29	D4 "	11	"	"	"	"
F30	D5 "	9	"	"	"	"
F31	D6 "	8	"	"	"	"
F32	D7 "	7	"	"	"	"

1/ Make connection at point indicated in table X.

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TABLE VIII. Output voltage (high-level) test.

Test condition	Parameter	Connect 1/	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
G1	BA0 Output voltage, high level	36	2.4	4.5	V	8
G2	BA1 "	37	"	"	"	"
G3	BA2 "	38	"	"	"	"
G4	BA3 "	39	"	"	"	"
G5	BA4 "	40	"	"	"	"
G6	BA5 "	20	"	"	"	"
G7	BA6 "	19	"	"	"	"
G8	BA7 "	18	"	"	"	"
G9	PB0 "	23	"	"	"	"
G10	PB1 "	22	"	"	"	"
G11	PB2 "	21	"	"	"	"
G12	PB3 "	2	"	"	"	"
G13	PB4 "	3	"	"	"	"
G14	PB5 "	4	"	"	"	"
G15	PB6 "	5	"	"	"	"
G16	PB7 "	6	"	"	"	"
G17	PC0 "	27	"	"	"	"
G18	PC1 "	26	"	"	"	"
G19	PC2 "	25	"	"	"	"
G20	PC3 "	24	"	"	"	"
G21	PC4 "	28	"	"	"	"

See footnote at end of table.

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TABLE VIII. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect 1/	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
G22	PC5 Output voltage, high level	29	2.4	4.5	V	8
G23	PC6 "	30	"	"	"	"
G24	PC7 "	31	"	"	"	"
G25	D0 "	15	"	"	"	"
G26	D1 "	14	"	"	"	"
G27	D2 "	13	"	"	"	"
G28	D3 "	12	"	"	"	"
G29	D4 "	11	"	"	"	"
G30	D5 "	9	"	"	"	"
G31	D6 "	8	"	"	"	"
G32	D7 "	7	"	"	"	"

1/ Make connection at point indicated in table XI.

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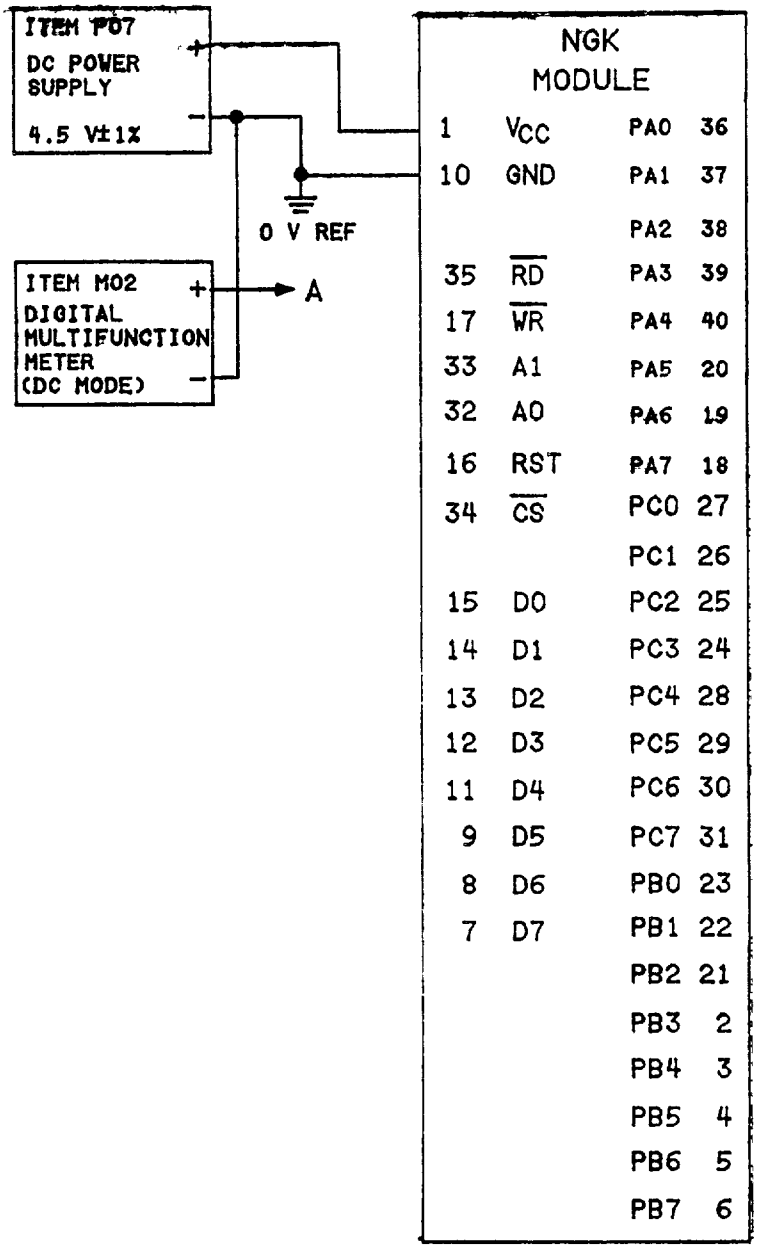


FIGURE 8. Output voltage (low-level and high-level) test circuit for tables VII and VIII.

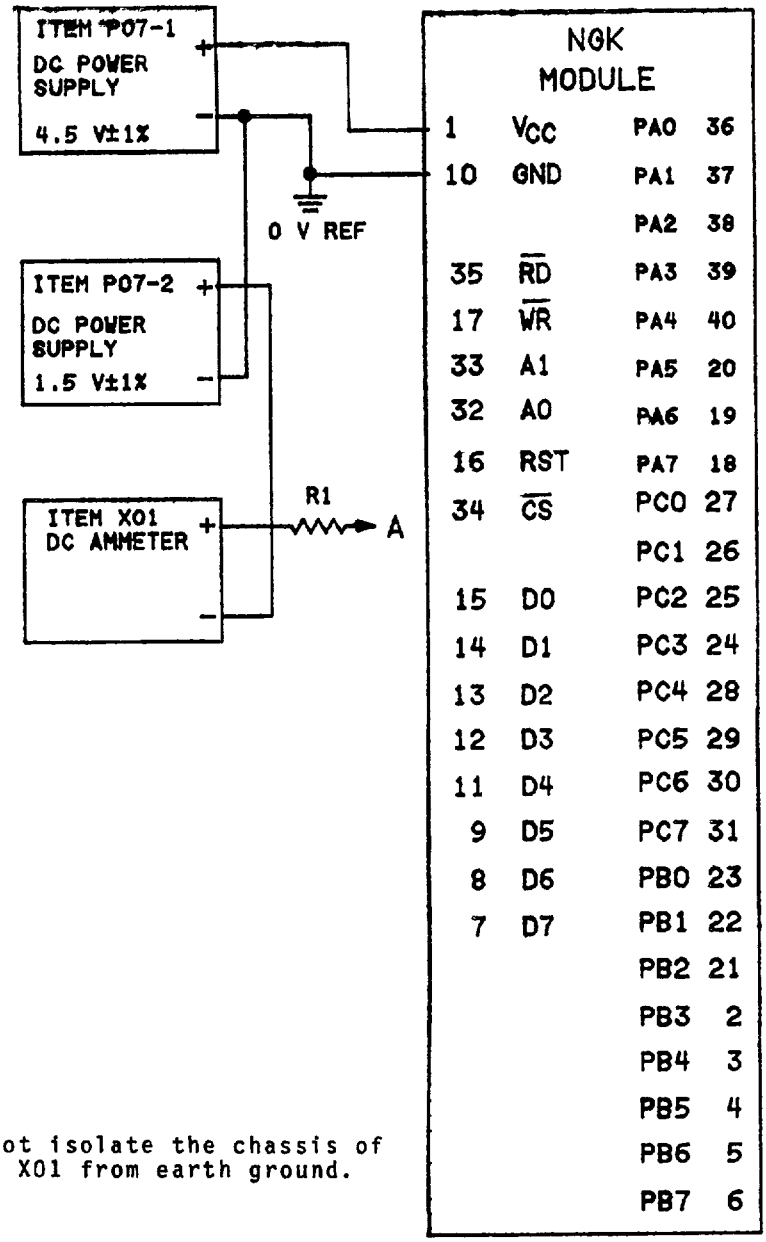
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TABLE IX. Output current (high-level, test.

Test condition	Parameter	Connect A to pin 1/	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
H1	PB0 Output current, high level	23 2/	1	4	mA	9
H2	PB1 "	22 "	"	"	"	"
H3	PB2 "	21 "	"	"	"	"
H4	PB3 "	2 "	"	"	"	"
H5	PB4 "	3 "	"	"	"	"
H6	PB5 "	4 "	"	"	"	"
H7	PB6 "	5 "	"	"	"	"
H8	PB7 "	6 "	"	"	"	"
H9	PC0 "	27 3/	"	"	"	"
H10	PC1 "	26 "	"	"	"	"
H11	PC2 "	25 "	"	"	"	"
H12	PC3 "	24 "	"	"	"	"
H13	PC4 "	28 "	"	"	"	"
H14	PC5 "	29 "	"	"	"	"
H15	PC6 "	30 "	"	"	"	"
H16	PC7 "	31 "	"	"	"	"

1/ Make connection at point indicated in table XI.
2/ Make test conditions H1-H8 measurements at step 10 of table XI.
3/ Make test conditions H9-H16 measurements at step 13 of table XI.

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CAUTION: Do not isolate the chassis of item X01 from earth ground.

NOTE: R1 = 750Ω ±1%, 1/2 W.

FIGURE 9. Output current (high-level) test circuit for table IX.

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TABLE X. Sequencing for low level output test.

Step no.	Control inputs						Data inputs D0-D7								Port conditions 1/
	33	32	35	17	34	16	7	8	9	11	12	13	14	15	
1	H	H	H	L	L	H	H	L	L	L	L	L	L	L	None
2	H	H	H	L	L	L	H	L	L	L	L	L	L	L	"
3	H	H	H	L	L	L	H	L	L	L	L	L	L	L	"
4	H	H	H	H	L	L	H	L	L	L	L	L	L	L	"
5	L	L	H	H	L	L	L	L	L	L	L	L	L	L	"
6	L	L	H	L	L	L	L	L	L	L	L	L	L	L	"
7	L	L	H	H	L	L	L	L	L	L	L	L	L	L	Measure port A
8	L	H	H	H	L	L	L	L	L	L	L	L	L	L	None
9	L	H	H	L	L	L	L	L	L	L	L	L	L	L	None
10	L	H	H	H	L	L	L	L	L	L	L	L	L	L	Measure port B
11	H	L	H	H	L	L	L	L	L	L	L	L	L	L	None
12	H	L	H	L	L	L	L	L	L	L	L	L	L	L	None
13	H	L	H	H	L	L	L	L	L	L	L	L	L	L	Measure port C
14	H	L	H	H	L	L	H	L	L	H	L	L	L	L	None
15	H	H	H	L	L	L	H	L	L	H	L	L	L	L	Port A inputs
16	L	L	H	H	L	L	← Open →								18 19 20 40 39 38 37 36
17	L	L	L	H	L	L	← Open →								L L L L L L L L
18	L	L	L	H	L	L	Measure D0-D7								L L L L L L L L

1/ Measurements are taken at step numbers 7, 10, and 13. Port A has forced logic levels at steps 17 and 18. The ports do not require any other conditioning and/or measurements.

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TABLE XI. Sequencing for high level output test.

Step no.	Control inputs						Data inputs D0-D7								Port conditions 1/
	33	32	35	17	34	16	7	8	9	11	12	13	14	15	
1	H	H	H	L	L	H	H	L	L	L	L	L	L	L	None
2	H	H	H	L	L	L	H	L	L	L	L	L	L	L	"
3	H	H	H	L	L	L	H	L	L	L	L	L	L	L	"
4	H	H	H	H	L	L	H	L	L	L	L	L	L	L	"
5	L	L	H	H	L	L	H	H	H	H	H	H	H	H	"
6	L	L	H	L	L	L	H	H	H	H	H	H	H	H	"
7	L	L	H	H	L	L	H	H	H	H	H	H	H	H	Measure port A
8	L	H	H	H	L	L	H	H	H	H	H	H	H	H	None
9	L	H	H	L	L	L	H	H	H	H	H	H	H	H	None
10	L	H	H	H	L	L	H	H	H	H	H	H	H	H	Measure port B
11	H	L	H	H	L	L	H	H	H	H	H	H	H	H	None
12	H	L	H	L	L	L	H	H	H	H	H	H	H	H	None
13	H	L	H	H	L	L	H	H	H	H	H	H	H	H	Measure port C
14	H	L	H	H	L	L	H	L	L	H	L	L	L	L	None
15	H	H	H	L	L	L	H	L	L	H	L	L	L	L	Port A inputs
16	L	L	H	H	L	L	← Open →								18 19 20 40 39 38 37 36
17	L	L	L	H	L	L	← Open →								H H H H H H H H
18	L	L	L	H	L	L	Measure D0-D7								H H H H H H H H

1/ Measurements are taken at step numbers 7, 10, and 13. Port A has forced logic levels at steps 17 and 18. The ports do not require any other conditioning and/or measurements.

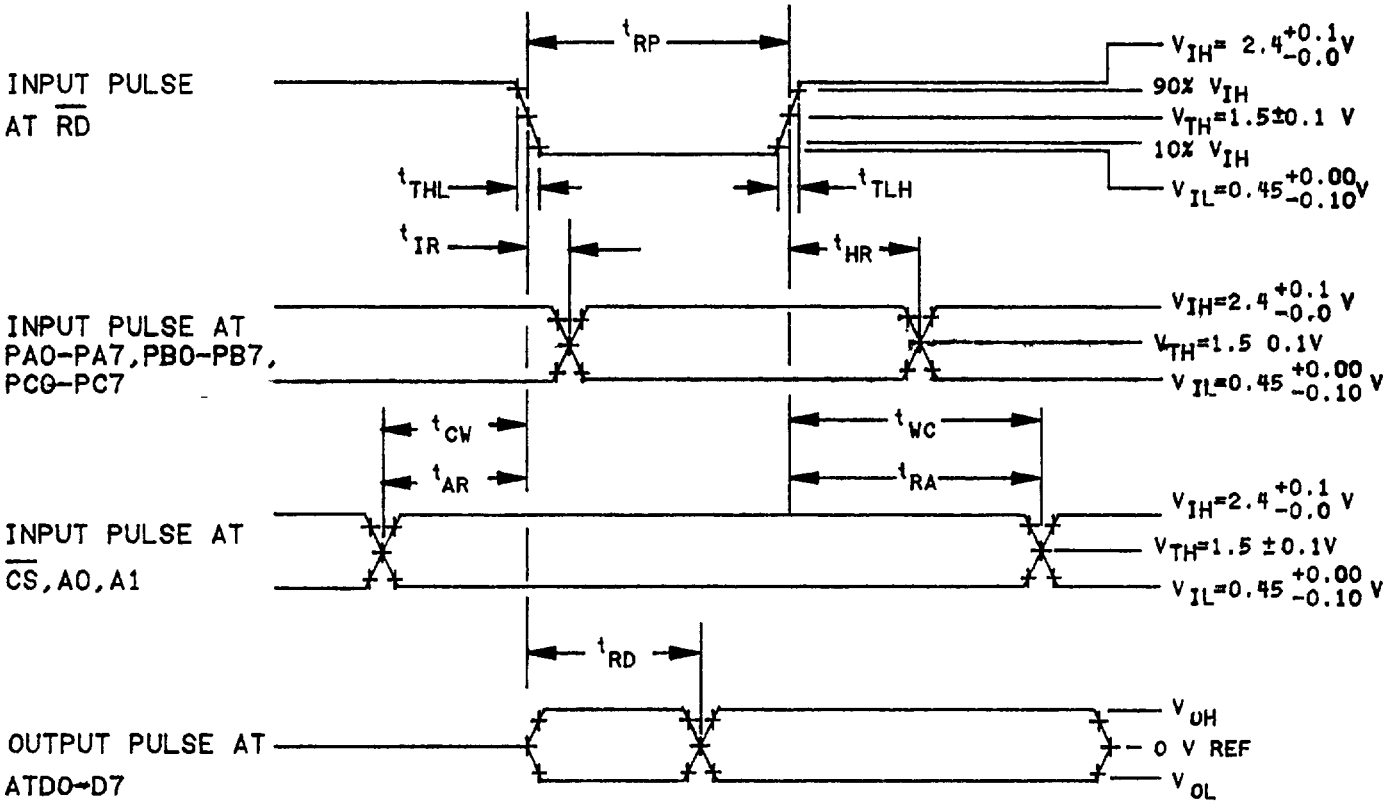
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TABLE XII. Dynamic functional test. 1/

Test condition	Parameter	Description	Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
J1	Dynamic functional	Functional test, minimum setup and hold time, and propagation delay time check, V_{CC} = 4.5 V	---	---	---	10,11,12, 13,14
J2	Dynamic functional	Functional test, minimum setup and hold time, and propagation delay time check, V_{CC} = 5.5 V	---	---	---	10,11,12, 13,14

1/ See paragraph 6.6 for additional information on the functional test vectors set.

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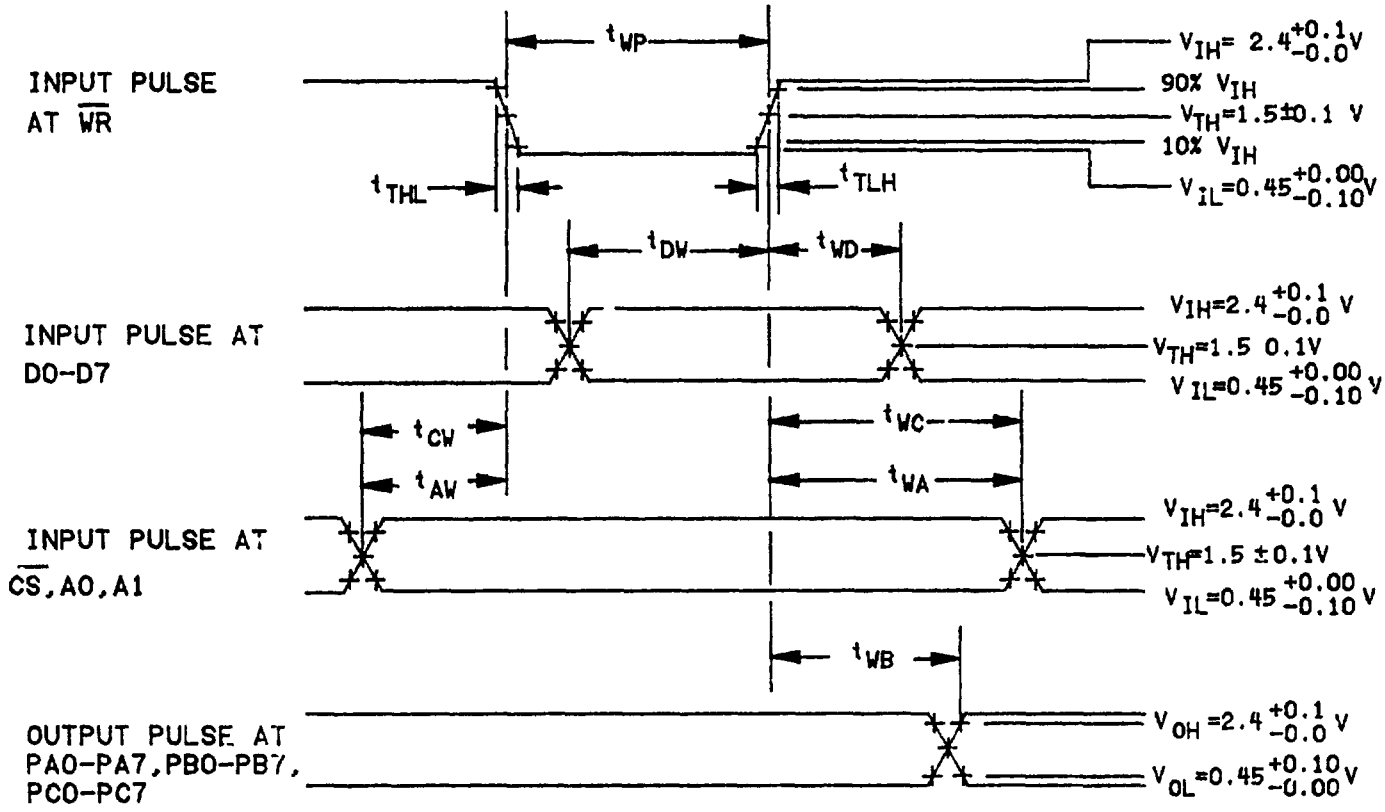
NOTES:

1. Input pulse parameters:
   Cycle time (t_{RW}) = 850 +8,-0 ns.
   Timing parameters = see table XIII.
   $t_{TLH} = t_{THL}$ = 5 ±5 ns.

2. Input parameters are set to the indicated values +2,-0 ns. Output parameters are the minimum time to start comparing outputs. The comparison extends for 10 ±2 ns.

FIGURE 10. Dynamic functional (basic input mode 0) test circuit waveforms for table XII.

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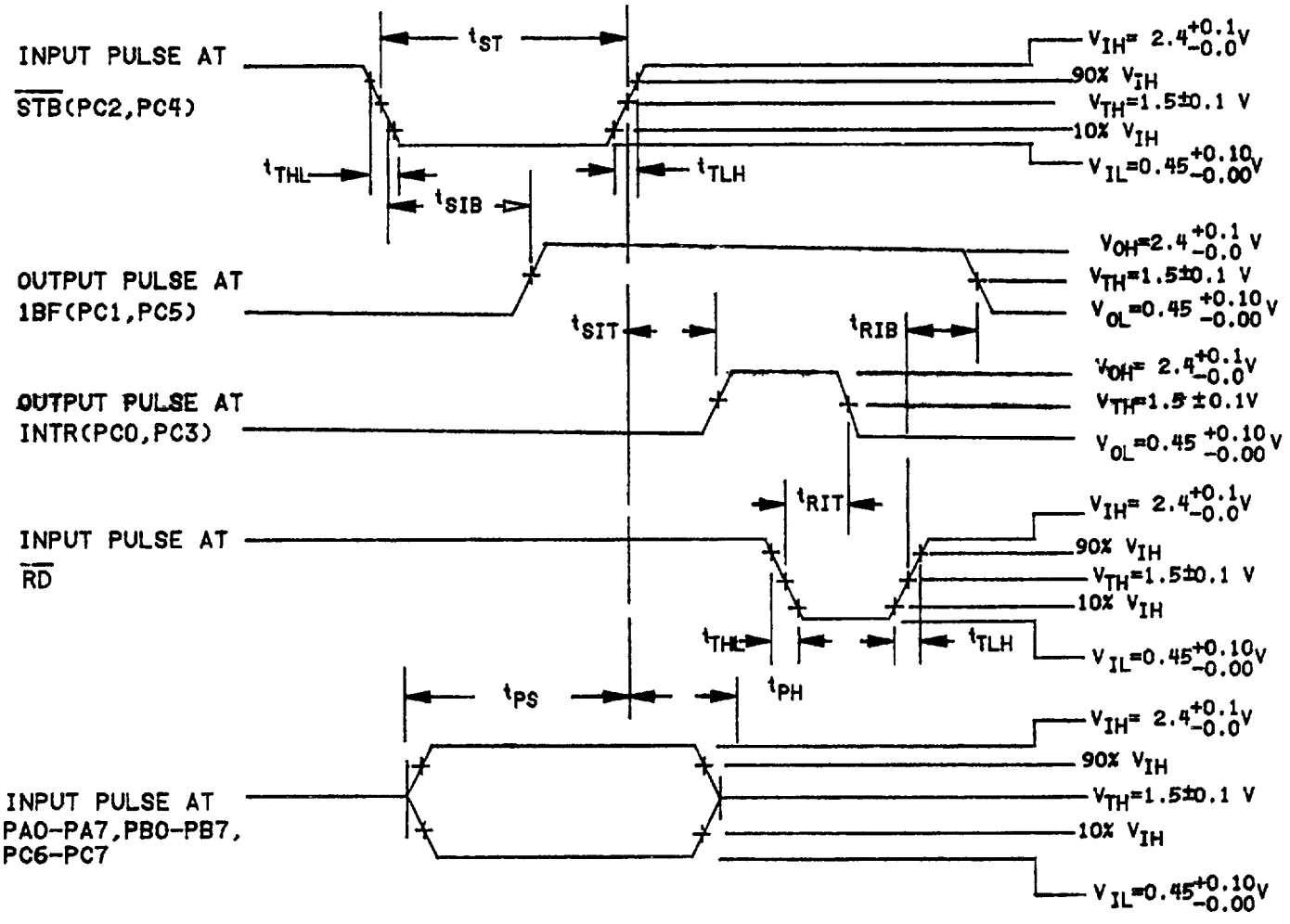
NOTES:

1. Input pulse parameters:
   Cycle time (t_{RW}) = 850 +8,-0 ns.
   Timing parameters = see table XIII.
   $t_{TLH} = t_{THL}$ = 5 ±5 ns.

2. Input parameters are set to the indicated values +2,-0 ns. Output parameters are the minimum time to start comparing outputs. The comparison extends for 10 ±2 ns.

FIGURE 11. Dynamic functional (basic output mode 0) test circuit waveforms for table XII.

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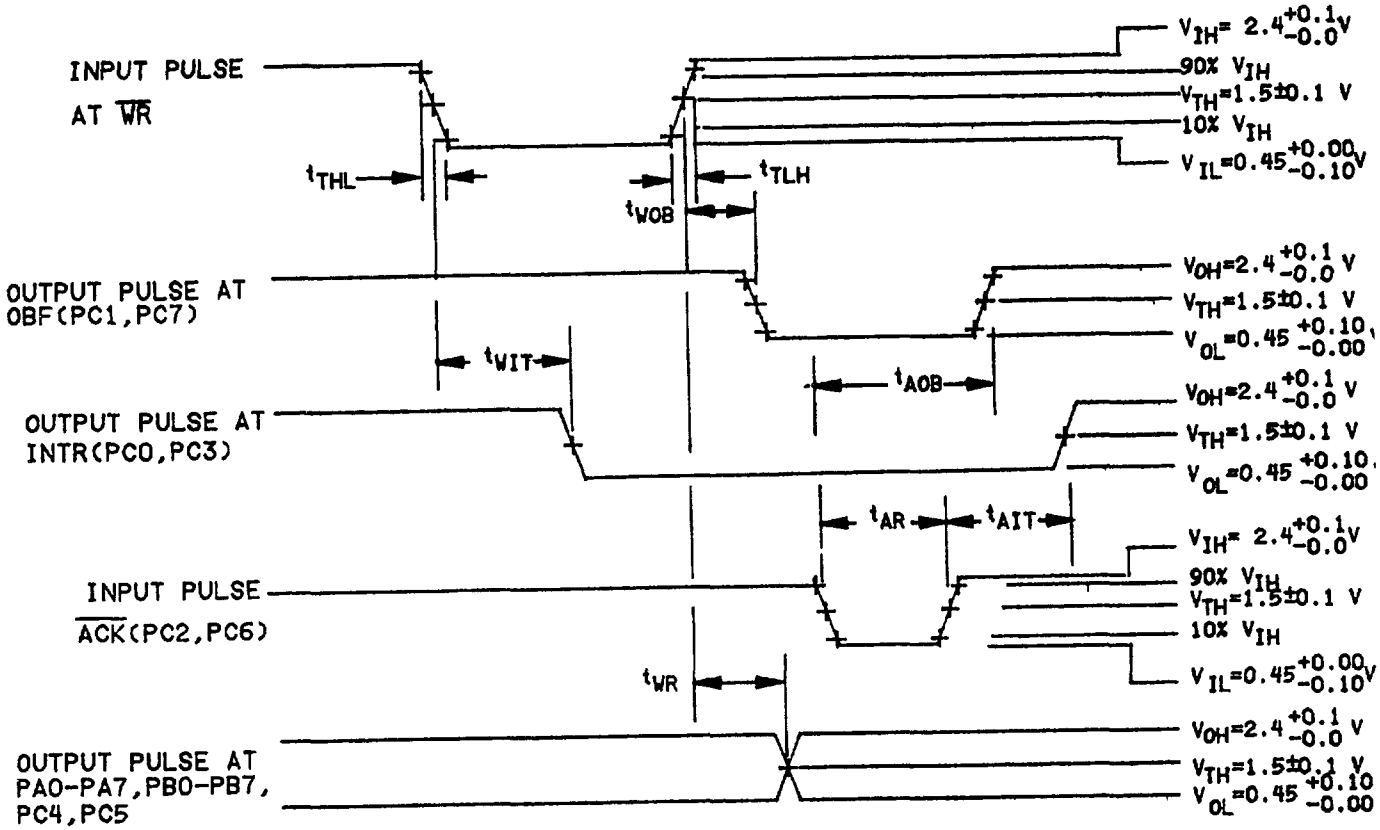
NOTES:

1. Input pulse parameters:
   Cycle time (t_{RW}) = 850 +8,-0 ns.
   Timing parameters = see table XIII.
   $t_{TLH} = t_{THL}$ = 5 ±5 ns.

2. Input parameters are set to the indicated values +2,-0 ns.
   Output parameters are the minimum time to start comparing outputs. The comparison extends for 10 ±2 ns.

FIGURE 12. Dynamic functional (strobed input, mode 1) test circuit waveforms for table XII.

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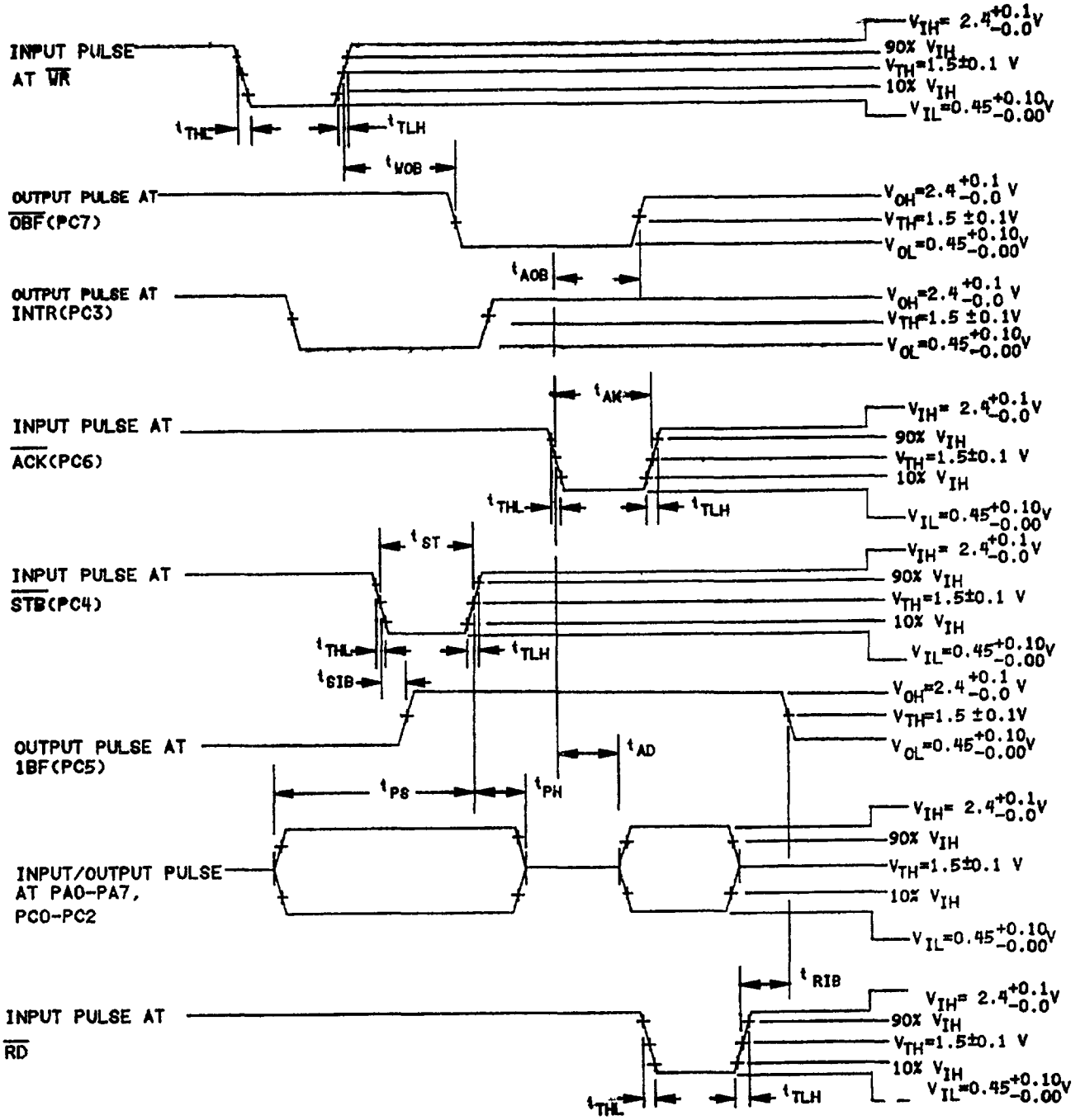
NOTES:

1. Input pulse parameters:
   Cycle time (t_{RW}) = 850 +8,-0 ns.
   Timing parameters = see table XIII.
   $t_{TLH} = t_{THL}$ = 5 ±5 ns.

2. Input parameters are set to the indicated values +2,-0 ns. Output parameters are the minimum time to start comparing outputs. The comparison extends for 10 ±2 ns.

FIGURE 13. Dynamic functional (strobed output, mode 1) test circuit waveforms for table XII.

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NOTES:

1. Input pulse parameters:
   Cycle time (t_{RW}) = 850 +8,-0 ns.
   Timing parameters = see table XIII.
   $t_{TLH} = t_{THL}$ = 5 ±5 ns.

2. Input parameters are set to the indicated values +2,-0 ns.
   Output parameters are the minimum time to start comparing outputs. The comparison extends for 10 ±2 ns.

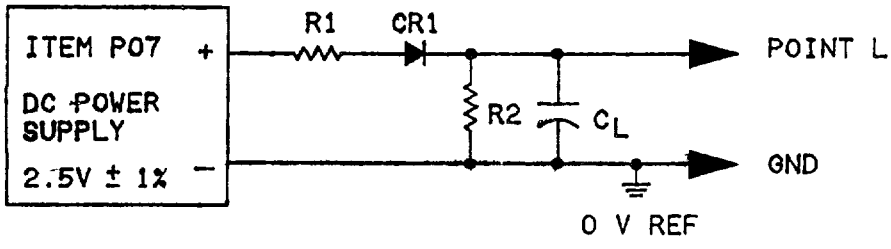
FIGURE 14. Dynamic functional (bidirectional, mode 2) test circuit waveforms for table XII.

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TABLE XIII. Setup and hold time, pulse width, cycle time, and propagation delay time parameters.

Parameter	Maximum limit
Setup times	
t_{DW} Data bus to $\overline{WR}$	100 ns
t_{AW} Address to $\overline{WR}$	0 ns
t_{CW} $\overline{CS}$ to $\overline{WR}$	20 ns
t_{IR} Peripheral bus to $\overline{RD}$	0 ns
t_{AR} Address to $\overline{RD}$	0 ns
t_{CR} $\overline{CS}$ to $\overline{RD}$	50 ns
t_{PS} Peripheral Bus to $\overline{STB}$ (PC2 or PC4)	0 ns
Hold times	
t_{WD} Data bus to $\overline{WR}$	30 ns
t_{WA} Address to $\overline{WR}$	20 ns
t_{WC} $\overline{CS}$ to $\overline{WR}$	35 ns
t_{HR} Peripheral bus to $\overline{RD}$	0 ns
t_{RA} Address to $\overline{RD}$	0 ns
t_{RC} $\overline{CS}$ to $\overline{RD}$	0 ns
t_{PH} Peripheral bus to $\overline{STB}$ (PC2 or PC4)	180 ns
Pulse widths and cycle times	
t_{RP} $\overline{RD}$	300 ns
t_{AK} $\overline{ACK}$ (PC2 or PC6)	300 ns
t_{ST} $\overline{STB}$ (PC2 or PC4)	500 ns
t_{RV} Time between read or writes	850 ns
t_{WP} $\overline{WR}$	400 ns
Propagation delay times	
t_{WB} $\overline{WR}$ to peripheral bus	350 ns
t_{RD} $\overline{RD}$ to data bus	250 ns
t_{AD} $\overline{ACK}$ (PC6) to peripheral bus	300 ns
t_{WOB} $\overline{WR}$ = 1 to $\overline{OBF}$ = 0 (PC1 or PC7)	650 ns
t_{AOB} $\overline{ACK}$ = 0 (PC2 or PC6) to $\overline{OBF}$ = 1 (PC1 or PC7)	350 ns
t_{SIB} $\overline{STB}$ = 0 (PC2 or PC4) to $\overline{IBF}$ = 1 (PC1 or PC5)	300 ns
t_{RIB} $\overline{RD}$ = 1 to IBF = 0 (PC1 or PC3)	300 ns
t_{RIT} $\overline{RD}$ = 0 to INTR = 0 (PC0 or PC3)	400 ns
t_{SIT} $\overline{STB}$ = 1 (PC2 or PC4) to INTR = 1 (PC0 or PC3)	300 ns
t_{AIT} $\overline{ACK}$ = 1 (PC2 or PC6) to INTR = 1 (PC0 or PC3)	350 ns
t_{WIT} $\overline{WR}$ = 0 to INTR = 0 (PC0 or PC3)	850 ns

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NOTES:

1. R1 = select for a current flow of 1.7 mA (794Ω) (PA0-PA7, PB0-PB7, PC0-PC7) and 2.5 mA (540Ω) (D0-D7) out of point L when point L is held at 0.45 V.
2. R2 = 12 kΩ ±1%, 1/2 W on PA0-PA7, PB0-PB7 and PC0-PC7.
   = 6 kΩ ±1%, 1/2 W on D0-D7.
3. CR1 = 1N4150 or equivalent.
4. C_L = 100 ±10 pF.

FIGURE 15. NGK logic load circuit.

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TABLE XIV. Disable delay time (t_{PHZ}) test.

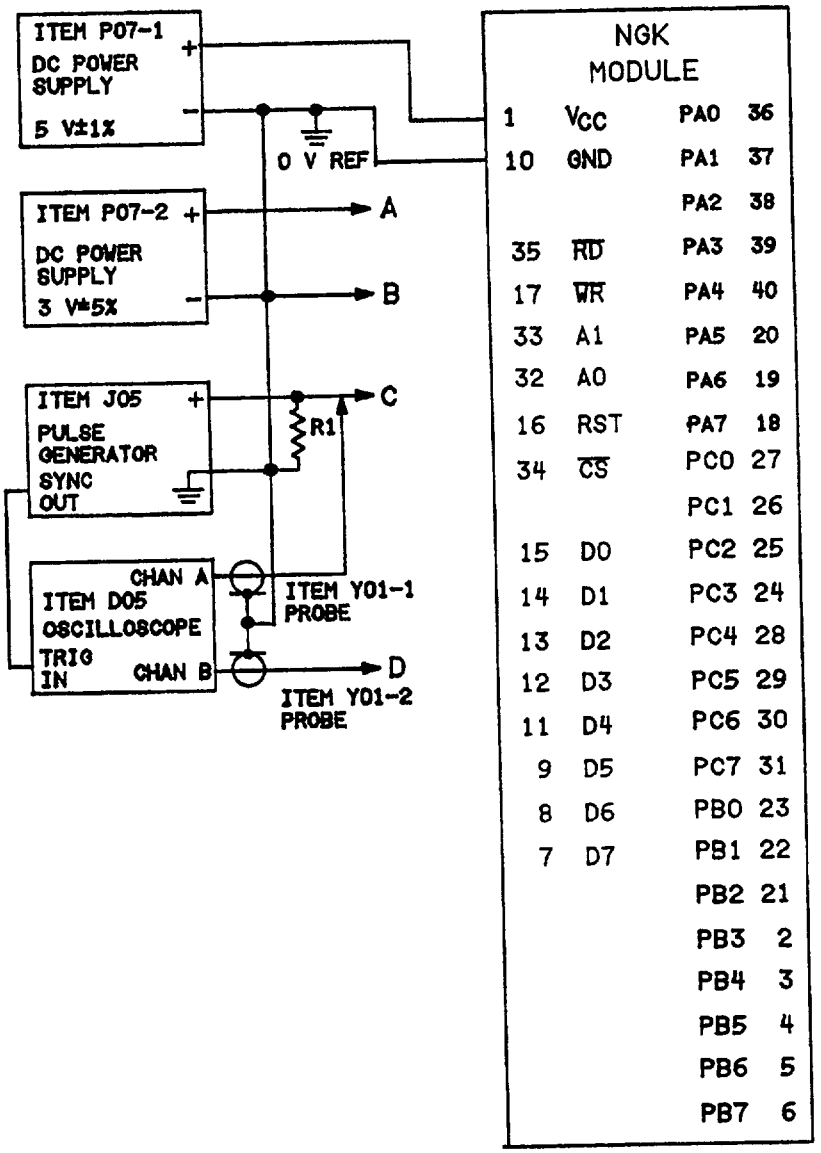
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	$\overline{RD}$ to D0 Disable delay time (t_{PHZ})				15	21	210	ns	16,17
K2	$\overline{RD}$ to D1 "				14	"	"	"	"
K3	$\overline{RD}$ to D2 "				13	"	"	"	"
K4	$\overline{RD}$ to D3 "				12	"	"	"	"
K5	$\overline{RD}$ to D4 "				11	"	"	"	"
K6	$\overline{RD}$ to D5 "				9	"	"	"	"
K7	$\overline{RD}$ to D6 "				8	"	"	"	"
K8	$\overline{RD}$ to D7 "				7	"	"	"	"
K9	$\overline{ACK}$ to PA0 "				36	40	400	"	"
K10	$\overline{ACK}$ to PA1 "				37	"	"	"	"
K11	$\overline{ACK}$ to PA2 "				38	"	"	"	"
K12	$\overline{ACK}$ to PA3 "				39	"	"	"	"
K13	$\overline{ACK}$ to PA4 "				40	"	"	"	"
K14	$\overline{ACK}$ to PA5 "				20	"	"	"	"
K15	$\overline{ACK}$ to PA6 "				19	"	"	"	"
K16	$\overline{ACK}$ to PA7 "				18	"	"	"	"

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TABLE XV. Disable delay time (t_{PLZ}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	$\overline{RD}$ to D0 Disable delay time (t_{PLZ})				15	10	150	ns	16,17
L2	$\overline{RD}$ to D1 "				14	"	"	"	"
L3	$\overline{RD}$ to D2 "				13	"	"	"	"
L4	$\overline{RD}$ to D3 "				12	"	"	"	"
L5	$\overline{RD}$ to D4 "				11	"	"	"	"
L6	$\overline{RD}$ to D5 "				9	"	"	"	"
L7	$\overline{RD}$ to D6 "				8	"	"	"	"
L8	$\overline{RD}$ to D7 "				7	"	"	"	"
L9	$\overline{ACK}$ to PA0 "				36	20	300	"	"
L10	$\overline{ACK}$ to PA1 "				37	"	"	"	"
L11	$\overline{ACK}$ to PA2 "				38	"	"	"	"
L12	$\overline{ACK}$ to PA3 "				39	"	"	"	"
L13	$\overline{ACK}$ to PA4 "				40	"	"	"	"
L14	$\overline{ACK}$ to PA5 "				20	"	"	"	"
L15	$\overline{ACK}$ to PA6 "				19	"	"	"	"
L16	$\overline{ACK}$ to PA7 "				18	"	"	"	"

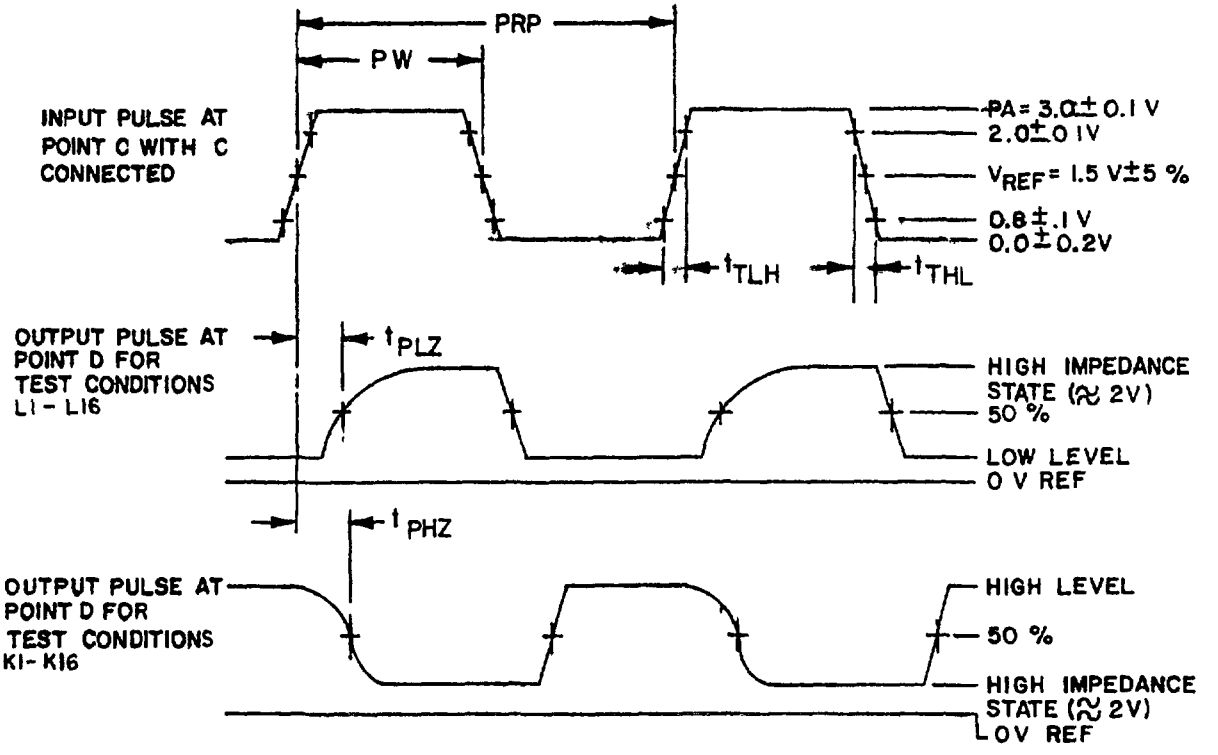
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NOTES:
1. R1 = 50Ω ±5%, 1/2 W.
2. Load outputs with load as shown on figure 15.
3. See table XVI for disable delay time sequencing for test conditions K1-K8 and L1-L8 and table XVII for test conditions K9-K16 and L9-L16.

FIGURE 16. Disable delay time (t_{PHZ} and t_{PLZ}) test circuit for tables XIV and XV.

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NOTES:
1. Input pulse parameters:
   PW = 1,000 ±100 ns.
   PRP = 2,000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making measurements, verify that the output high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltage levels are referenced to ground.

FIGURE 17. Disable delay time (t_{PHZ} and t_{PLZ}) test circuit waveforms for tables XIV and XV.

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TABLE XVI. Sequencing for disable delay time test. 1/

Step no.	Control inputs						Data bus pins								Port A pins							
	33	32	34	35	17	16	7	8	9	11	12	13	14	15	18	19	20	40	39	38	37	36
1	H	H	L	H	H	H	L	L	L	H	L	L	L	L	← Open →							
2	H	H	L	H	H	L	L	L	L	H	L	L	L	L	"							
3	H	H	L	H	L	L	L	L	L	H	L	L	L	L	"							
4	H	H	L	H	H	L	L	L	L	H	L	L	L	L	"							
5	L	L	L	H	H	L	← Open →								H	H	H	H	H	H	H	H
6	L	L	L	P	H	L	Measure K1-K8								H	H	H	H	H	H	H	H
7	L	L	L	P	H	L	Measure L1-L8								L	L	L	L	L	L	L	L

1/ L = 0 V ±5%.
H = 3 V ±5%.
P = pulse as shown on figure 17.

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TABLE XVII. Sequencing for disable delay time test. 1/

Step no.	Control pins							Data bus pins								Port A pins							
	30	33	32	34	35	17	16	7	8	9	11	12	13	14	15	18	19	20	40	39	38	37	36
1	O	H	H	L	H	L	H	H	H	L	L	L	L	L	L	← Open →							
2	O	H	H	L	H	L	L	H	H	L	L	L	L	L	L	"							
3	O	H	H	L	H	H	L	H	H	L	L	L	L	L	L	"							
4	P	L	L	L	H	L	L	H	H	H	H	H	H	H	H	"							
5	P	L	L	L	H	H	L	H	H	H	H	H	H	H	H	Measure K9-K16							
6	P	L	L	L	H	H	L	L	L	L	L	L	L	L	L	Open							
7	P	L	L	L	H	L	L	L	L	L	L	L	L	L	L	Open							
8	P	L	L	L	H	H	L	L	L	L	L	L	L	L	L	Measure L9-L16							

1/ L = 0 V ±5%.
H = 3 V ±5%.
P = Pulse as shown on figure 17.
O = Open.

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TABLE XVIII. Power supply transient amplitude test.

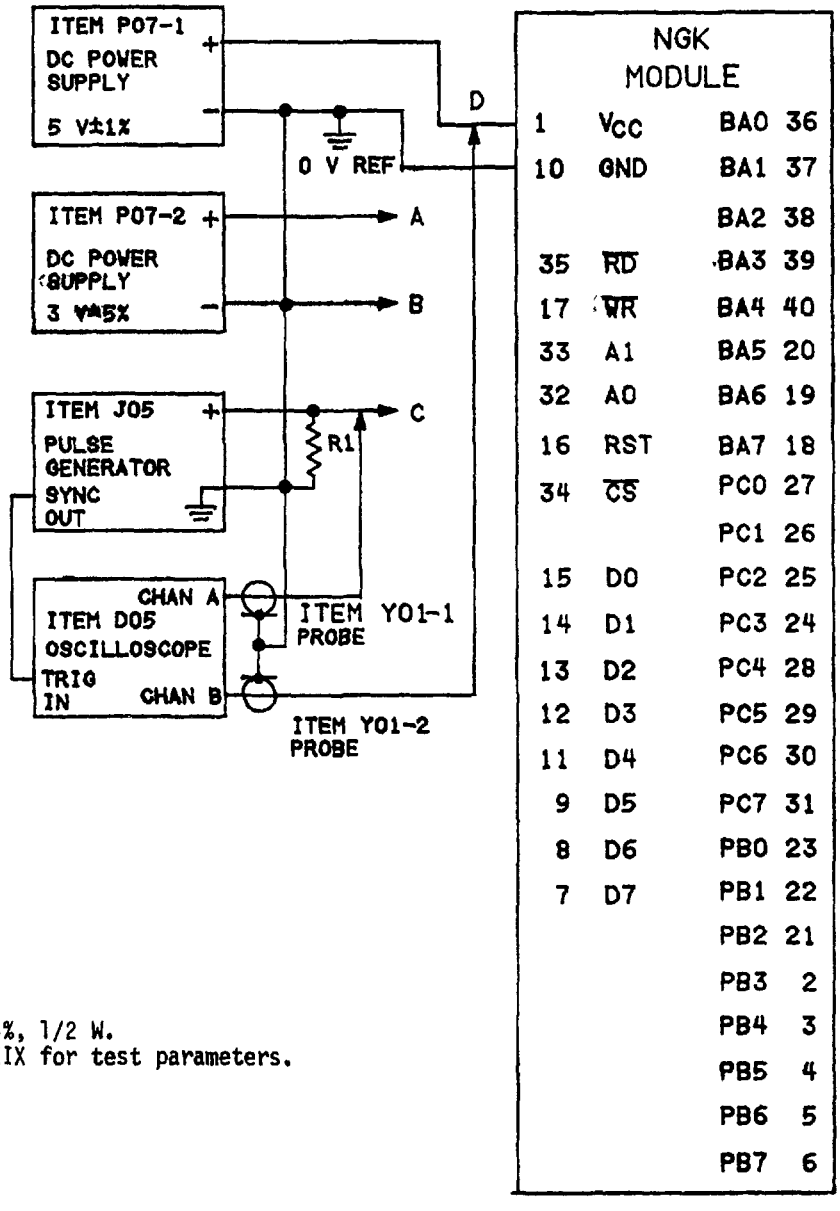
Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply transient amplitdue	0	400	mV	18,19

TABLE XIX. Power supply transient amplitude sequencing test. 1/

Step no.	Control pins						Data bus pins								Port A pins							
	33	32	34	35	17	16	7	8	9	11	12	13	14	15	18	19	20	40	39	38	37	36
1	H	H	L	H	H	H	L	L	L	H	L	L	L	L	← Open →							
2	H	H	L	H	L	L	L	L	L	H	L	L	L	L								
3	H	H	L	H	H	L	L	L	L	H	L	L	L	L								
4	L	L	L	H	H	L	← Open →															
5	L	L	L	L	H	L	← Open →															
6	L	L	L	L	H	L	← Open →								Pulse to PA0-PA7 Measure PSTA							

1/ L = 0.0 ±0.1 V.
H = 3 V ± 5%.
P = Pulse as shown on figure 19.

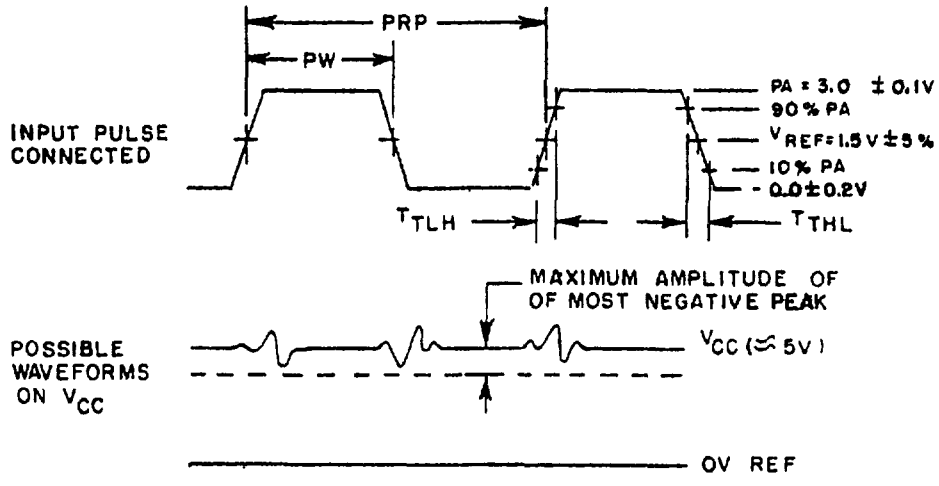
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NOTES:
1. R1 = 50Ω ±5%, 1/2 W.
2. See table XIX for test parameters.

FIGURE 18. Power supply transient amplitude test circuit for table XVIII.

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NOTE: Input pulse parameters.
PW = 500 ±50 ns.
PRP = 1,000 ±100 ns.
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.

FIGURE 19. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XX. Output crosstalk (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		B to pin	C to pin	D to pin	Min	Max		
N1	PA0 Output crosstalk, low level	15	14,13,12,11,9,8,7	36	0	400	mV	20,21
N2	PA1 "	14	15,13,12,11,9,8,7	37	"	"	"	"
N3	PA2 "	13	15,14,12,11,9,8,7	38	"	"	"	"
N4	PA3 "	12	15,14,13,11,9,8,7	39	"	"	"	"
N5	PA4 "	11	15,14,13,12,9,8,7	40	"	"	"	"
N6	PA5 "	9	15,14,13,12,11,8,7	20	"	"	"	"
N7	PA6 "	8	15,14,13,12,11,9,7	19	"	"	"	"
N8	PA7 "	7	15,14,13,12,11,9,8	18	"	"	"	"
N9	PB0 "	15	14,13,12,11,9,8,7	23	"	"	"	"
N10	PB1 "	14	15,13,12,11,9,8,7	22	"	"	"	"
N11	PB2 "	13	15,14,12,11,9,8,7	21	"	"	"	"
N12	PB3 "	12	15,14,13,11,9,8,7	2	"	"	"	"
N13	PB4 "	11	15,14,13,12,9,8,7	3	"	"	"	"
N14	PB5 "	9	15,14,13,12,11,8,7	4	"	"	"	"
N15	PB6 "	8	15,14,13,12,11,9,7	5	"	"	"	"
N16	PB7 "	7	15,14,13,12,11,9,8	6	"	"	"	"
N17	PC0 "	15	14,13,12,11,9,8,7	27	"	"	"	"
N18	PC1 "	14	15,13,12,11,9,8,7	26	"	"	"	"
N19	PC2 "	13	15,14,12,11,9,8,7	25	"	"	"	"
N20	PC3 "	12	15,14,13,11,9,8,7	24	"	"	"	"
N21	PC4 "	11	15,14,13,12,9,8,7	28	"	"	"	"

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TABLE XX. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		B to pin	C to pin	D to pin	Min	Max		
N22	PC5 Output crosstalk, low level	9	15,14,13,12,11,8,7	29	0	400	mV	20,21
N23	PC6 "	8	15,14,13,12,11,9,7	30	"	"	"	"
N24	PC7 "	7	15,14,13,12,11,9,8	31	"	"	"	"
N25	D0 "	36	37,38,39,40,20,19,18	15	"	"	"	"
N26	D1 "	37	36,38,39,40,20,19,18	14	"	"	"	"
N27	D2 "	38	36,37,39,40,20,19,18	13	"	"	"	"
N28	D3 "	39	36,37,38,40,20,19,18	12	"	"	"	"
N29	D4 "	40	36,37,38,39,20,19,18	11	"	"	"	"
N30	D5 "	20	36,37,38,39,40,19,18	9	"	"	"	"
N31	D6 "	19	36,37,38,39,40,20,18	8	"	"	"	"
N32	D7 "	18	36,37,38,39,40,20,19	7	"	"	"	"

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TABLE XXI. Output crosstalk (high-level) test.

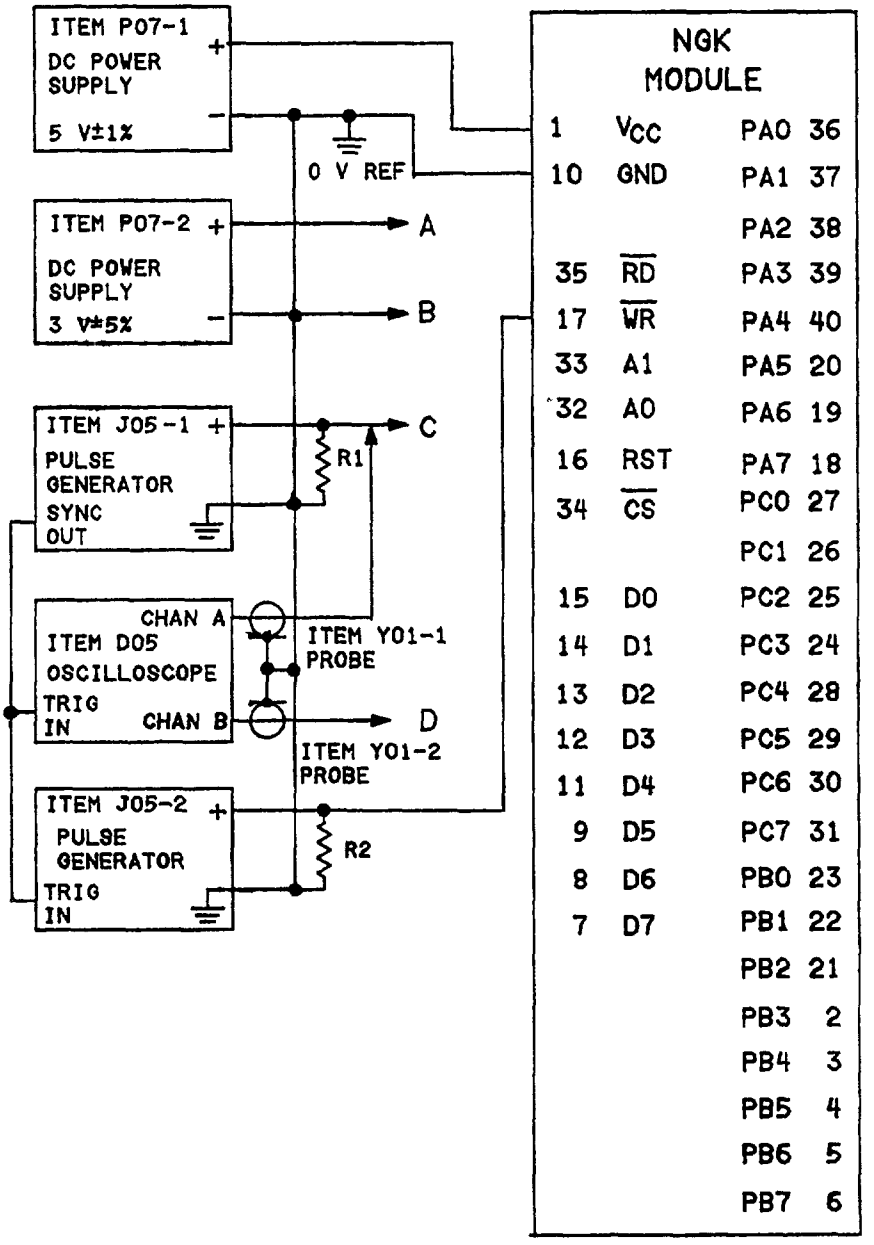
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		B to pin	C to pin	D to pin	Min	Max		
P1	PA0 Output crosstalk, high level	15	14,13,12,11,9,8,7	36	0	400	mV	20,21
P2	PA1 "	14	15,13,12,11,9,8,7	37	"	"	"	"
P3	PA2 "	13	15,14,12,11,9,8,7	38	"	"	"	"
P4	PA3 "	12	15,14,13,11,9,8,7	39	"	"	"	"
P5	PA4 "	11	15,14,13,12,9,8,7	40	"	"	"	"
P6	PA5 "	9	15,14,13,12,11,8,7	20	"	"	"	"
P7	PA6 "	8	15,14,13,12,11,9,7	19	"	"	"	"
P8	PA7 "	7	15,14,13,12,11,9,8	18	"	"	"	"
P9	PB0 "	15	14,13,12,11,9,8,7	23	"	"	"	"
P10	PB1 "	14	15,13,12,11,9,8,7	22	"	"	"	"
P11	PB2 "	13	15,14,12,11,9,8,7	21	"	"	"	"
P12	PB3 "	12	15,14,13,11,9,8,7	2	"	"	"	"
P13	PB4 "	11	15,14,13,12,9,8,7	3	"	"	"	"
P14	PB5 "	9	15,14,13,12,11,8,7	4	"	"	"	"
P15	PB6 "	8	15,14,13,12,11,9,7	5	"	"	"	"
P16	PB7 "	7	15,14,13,12,11,9,8	6	"	"	"	"
P17	PC0 "	15	14,13,12,11,9,8,7	27	"	"	"	"
P18	PC1 "	14	15,13,12,11,9,8,7	26	"	"	"	"
P19	PC2 "	13	15,14,12,11,9,8,7	25	"	"	"	"
P20	PC3 "	12	15,14,13,11,9,8,7	24	"	"	"	"
P21	PC4 "	11	15,14,13,12,9,8,7	28	"	"	"	"

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TABLE XXI. Output crosstalk (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		B to pin	C to pin	D to pin	Min	Max		
P22	PC5 Output crosstalk, high level	9	15,14,13,12,11,8,7	29	0	400	mV	20,21
P23	PC6 "	8	15,14,13,12,11,9,7	30	"	"	"	"
P24	PC7 "	7	15,14,13,12,11,9,8	31	"	"	"	"
P25	D0 "	36	37,38,39,40,20,19,18	15	"	"	"	"
P26	D1 "	37	36,38,39,40,20,19,18	14	"	"	"	"
P27	D2 "	38	36,37,39,40,20,19,18	13	"	"	"	"
P28	D3 "	39	36,37,38,40,20,19,18	12	"	"	"	"
P29	D4 "	40	36,37,38,39,20,19,18	11	"	"	"	"
P30	D5 "	20	36,37,38,39,40,19,18	9	"	"	"	"
P31	D6 "	19	36,37,38,39,40,20,18	8	"	"	"	"
P32	D7 "	18	36,37,38,39,40,20,19	7	"	"	"	"

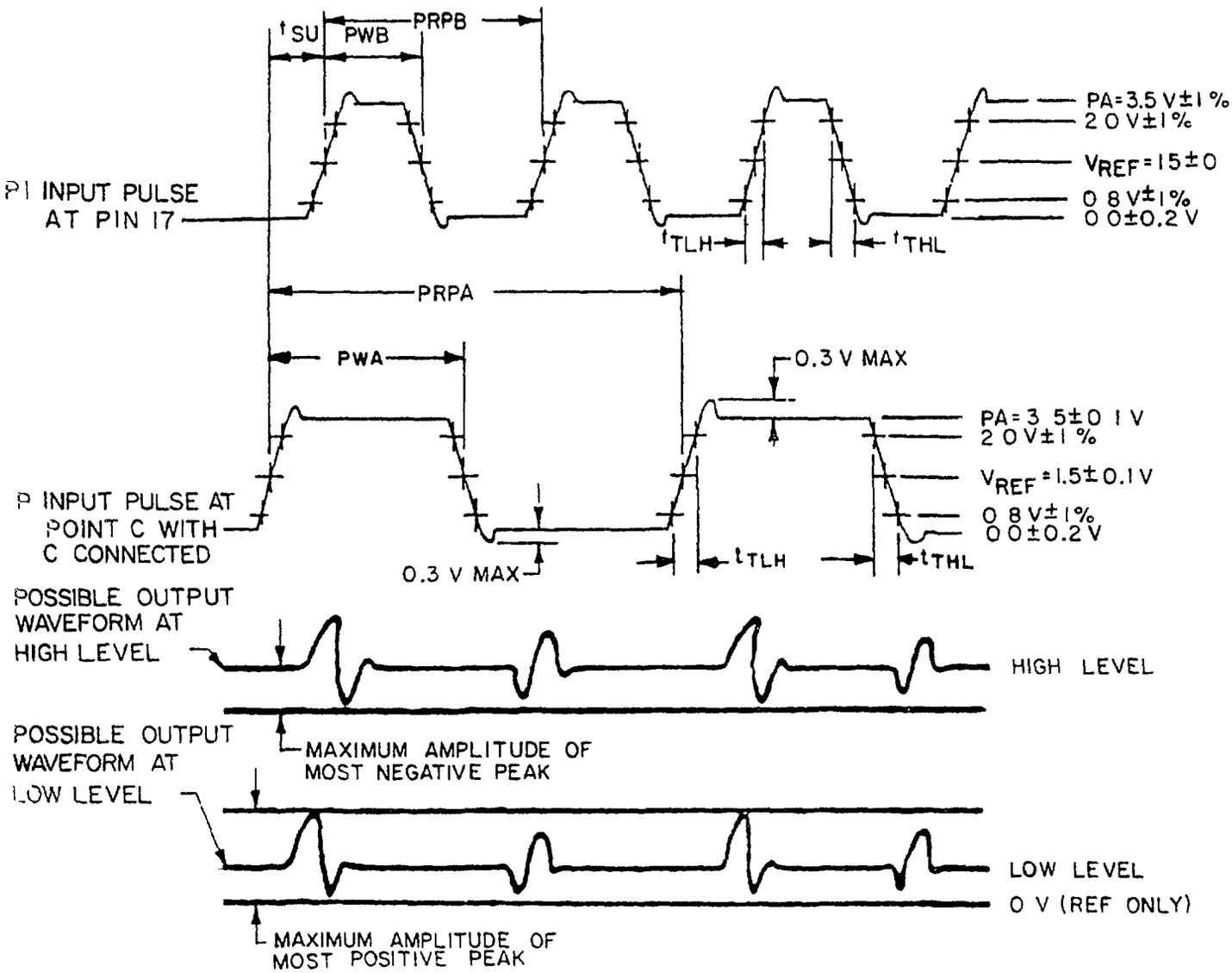
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1/2 W.
2. See table XXII for test parameters.
3. Load all outputs with load as shown on figure 15.

FIGURE 20. Output crosstalk (low-level and high-level) test circuit for tables XX and XXI.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at all input pins).
   PWA = 2.0 µs ±0.1 µs.
   PWB = 1 µs ±0.1 µs.
   t_{SU} = 500 ns.
   PRPA = 4.0 µs ±0.2 µs.
   PRPB = 2 µs ±0.2 µs.
2. The lead length of electrical conductors shall be less than 2 inches (50.8 mm) between resistor R1 and the input pins; items Y01-1 and Y01-2 and the input and output pins; load circuits and the output pins; and ground leads of the items Y01-1 and Y01-2 and the test fixture and module ground pins.

FIGURE 21. Output crosstalk (low-level and high-level) test circuit waveforms for tables XX and XXI.

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TABLE XXII. Output crosstalk (low-level and high-level) sequencing test. 1/

Step no.	Control inputs						Data inputs D0-D7								Port conditions							
	33	32	35	17	34	16	7	8	9	11	12	13	14	15								
1	H	H	H	L	L	H	H	L	L	L	L	L	L	L	None							
2	H	H	H	L	L	L	H	L	L	L	L	L	L	L	"							
3	H	H	H	L	L	L	H	L	L	L	L	L	L	L	"							
4	H	H	H	H	L	L	H	L	L	L	L	L	L	L	"							
5	L	L	H	P1	L	L	P	P	P	P	P	P	P	P	"							
6	L	L	H	P1	L	L	P	P	P	P	P	P	P	P	"							
7	L	L	H	P1	L	L	P	P	P	P	P	P	P	P	Measure port A							
8	L	H	H	P1	L	L	P	P	P	P	P	P	P	P	None							
9	L	H	H	P1	L	L	P	P	P	P	P	P	P	P	None							
10	L	H	H	P1	L	L	P	P	P	P	P	P	P	P	Measure port B							
11	H	L	H	P1	L	L	P	P	P	P	P	P	P	P	None							
12	H	L	H	P1	L	L	P	P	P	P	P	P	P	P	None							
13	H	L	H	P1	L	L	P	P	P	P	P	P	P	P	Measure port C							
14	H	L	H	H	L	L	H	L	L	H	L	L	L	L	None							
15	H	H	H	L	L	L	H	L	L	H	L	L	L	L	Port A inputs							
16	L	L	H	H	L	L	← Open →								18	19	20	40	39	38	37	36
17	L	L	L	H	L	L	← Open →								P	P	P	P	P	P	P	P
18	L	L	L	H	L	L	Measure D0-D7								P	P	P	P	P	P	P	P

1/ L = 0 V ± 5%.
H = 3 V ± 5%.
P = Pulse as shown on figure 21.
P1 = Pulse as shown on figure 21.

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TABLE XXIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		C to pin	D to pin	Min	Max		
Q	Power supply filter capacitance	1	10	1.5	2.9	µF	22

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TABLE XXIV. Input capacitance test.

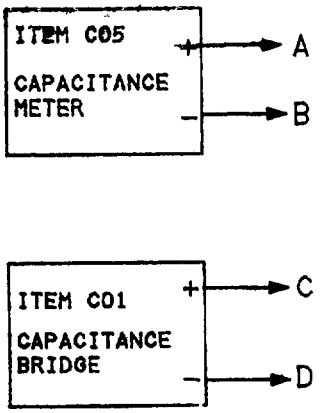
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	D0 Input capacitance	15	10	2	25	pF	22
R2	D1 "	14	"	"	"	"	"
R3	D2 "	13	"	"	"	"	"
R4	D3 "	12	"	"	"	"	"
R5	D4 "	11	"	"	"	"	"
R6	D5 "	9	"	"	"	"	"
R7	D6 "	8	"	"	"	"	"
R8	D7 "	7	"	"	"	"	"
R9	PA0 "	36	"	"	"	"	"
R10	PA1 "	37	"	"	"	"	"
R11	PA2 "	38	"	"	"	"	"
R12	PA3 "	39	"	"	"	"	"
R13	PA4 "	40	"	"	"	"	"
R14	PA5 "	20	"	"	"	"	"
R15	PA6 "	19	"	"	"	"	"
R16	PA7 "	18	"	"	"	"	"
R17	PB0 "	23	"	"	"	"	"
R18	PB1 "	22	"	"	"	"	"
R19	PB2 "	21	"	"	"	"	"
R20	PB3 "	2	"	"	"	"	"
R21	PB4 "	3	"	"	"	"	"

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TABLE XXIV. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R22	PB5 Input capacitance	4	10	2	25	pF	22
R23	PB6 "	5	"	"	"	"	"
R24	PB7 "	6	"	"	"	"	"
R25	PC0 "	27	"	"	"	"	"
R26	PC1 "	26	"	"	"	"	"
R27	PC2 "	25	"	"	"	"	"
R28	PC3 "	24	"	"	"	"	"
R29	PC4 "	28	"	"	"	"	"
R30	PC5 "	29	"	"	"	"	"
R31	PC6 "	30	"	"	"	"	"
R32	PC7 "	31	"	"	"	"	"
R33	$\overline{RD}$ "	35	"	"	15	"	"
R34	$\overline{WR}$ "	17	"	"	"	"	"
R35	A1 "	33	"	"	"	"	"
R36	A0 "	32	"	"	"	"	"
R37	RST "	16	"	"	"	"	"
R38	$\overline{CS}$ "	34	"	"	"	"	"

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NGK MODULE			
1	V_{CC}	PA0	36
10	GND	PA1	37
		PA2	38
35	$\overline{RD}$	PA3	39
17	$\overline{WR}$	PA4	40
33	A1	PA5	20
32	A0	PA6	19
16	RST	PA7	18
34	$\overline{CS}$	PC0	27
		PC1	26
15	D0	PC2	25
14	D1	PC3	24
13	D2	PC4	28
12	D3	PC5	29
11	D4	PC6	30
9	D5	PC7	31
8	D6	PB0	23
7	D7	PB1	22
		PB2	21
		PB3	2
		PB4	3
		PB5	4
		PB6	5
		PB7	6

NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 22. Power supply filter and input capacitance test circuit for tables XXIII and XXIV.

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